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REPORT

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Influenza between August 2003 and July 2004

Influenza A H3N2 viruses were principally responsible for outbreaks of influenza throughout the world. In western Europe and North America, influenza activity began in October, somewhat earlier and more severe than in the preceding 3 years. In general, influenza activity was mild to moderate throughout this period. Influenza A H1N1 and H1N2 viruses circulated at low levels in most parts of the world and caused outbreaks in only a few countries; an outbreak in Iceland during October and November was due mainly to AH1N2 viruses, whereas AH1N1 viruses were responsible for localized outbreaks in the United Kingdom in May. Influenza B viruses circulated at low levels.

A single case of human infection by an influenza A H9N2 virus was identified in Hong Kong SAR in December 2003. During January to March 2004, 34 human cases of AH5N1 infection, 23 of which were fatal, were associated with extensive outbreaks of highly pathogenic influenza A H5N1 in poultry in Vietnam and Thailand. Further cases have subsequently been associated with a resurgence of outbreaks in poultry since July. In March 2004, two cases of human infection by influenza A H7N3 viruses were associated with outbreaks of influenza A H7N3 in poultry in British Columbia, Canada.

Characteristics of the viruses

During the past year, the Influenza Centre characterized approximately 1100 human influenza viruses from some 44 countries. Ninety four per cent were influenza A, of which 90% were H3N2, 5% were H1N1 and 5% were H1N2, and 6% were influenza B (Table 1). The majority of viruses (AH3N2) were isolated during November to February (Table 2).

A H1N1 and H1N2 viruses

In haemaggutination inhibition (HI) tests using post-infection ferret antisera, the HAs of most of the H1N1 and H1N2 viruses were shown to be indistinguishable from each other and to be closely related to that of the current vaccine strain, A/New Caledonia/20/99 (H1N1), and A/Egypt/96/02 (H1N2) (Table 3).

The HA and NA sequences of many of the H1N1 viruses were shown to be similar to viruses isolated during the previous year (Figures 1 and 3). Some, e.g. A/England/40/04, possessed the additional amino acid change Y252F in HA (Figure 2). The NA sequences were characterized by the changes E332K and N450D relative to A/New Caledonia/20/99 (Figure 4, Table 4). A few, represented by A/Hong Kong/2765/04, formed a separate phylogenetic group and were characterized by 5 amino acid changes in HA, T82K, Y94H, R145K, R208K and T266N, and 3 changes in NA, E214G, V234M and D382N (Table 4).

The HA sequences of H1N2 viruses, like previous isolates, had 4 changes relative to the HA of A/New Caledonia/20/99, V165A, V174I, A189T and A214T. Viruses isolated in Senegal had the characteristic change G152E, whereas viruses responsible for the outbreak in Iceland together with viruses isolated in some other countries towards the end of 2003 were distinguished by 4 amino changes, K73R, G185R, R188M and Q282K, in HA (Figure 2). The NAs were also similar in sequence to those of viruses isolated during the previous year (Figure 7); additional amino acid changes common to recent isolates included E41G and S450T, or V30I and K308R, respectively (Figure 8, Table 4).

A H3N2 viruses
Most H3N2 viruses were antigenically closely related to A/Fujian/411/02 and the A/Fujian/411/02-like reference viruses; very few were A/Panama/2007/99-like (Table 5). Many gave reduced HI titres, relative to the homologous egg-grown virus, with antisera to the vaccine strain A/Wyoming/3/03, and appeared more closely related to A/Christchurch/28/03, which was representative in HA and NA sequences of viruses prevalent during the latter half of 2003 and early 2004 (see below). More recent HI tests including an antiserum to A/Wellington/1/04 (from A. Hampson) showed that recent isolates were in general closely related to A/Wellington/1/04.

HA sequences distinguished two phylogenetic groups (Figure 5). Group 1 containing the HAs of A/Fujian/411/02 and A/Wyoming/3/03 and the more recent variants represented by A/Wellington/1/04, with amino acid changes Y159F, S189N and S227P relative to A/Fujian/411/02, and by A/Shantou/1219/04, with the additional differences K145N(D) and V226I (Figures 5 and 6, Table 6). The NA sequences of the "A/Wellington/1/04 subgroup" were characterized by 3 changes relative to the NA of A/Fujian/411/02, N93D, E199K and Q432E, whereas those of the "A/Shantou/1219/04 subgroup" were characterized by the amino acid changes, E199K, I215V and K431N relative to the NA of A/Fujian/411/02 (Figures 7 and 8, Table 6). Group 2 viruses, represented by A/Christchurch/28/03, possess the change N126D which removes the glycosylation site; these viruses have been relatively less prevalent during mid to late 2004. Other changes in the HAs of more recent variants include S45I, G50E and N144D (causing loss of the glycosylation site) of e.g. A/Iceland/6/04, or A138S and K145N of e.g. A/Shantou/575/04. The NA sequences of a number of these viruses were shown to be similar to those of previous isolates within this group, which are closely related to the NAs of earlier A/New York/55/01-like (A/Panama/2007/99-like) viruses; they differ in 16/17 amino acids from the NA of A/Fujian/411/02 (Table 6).

Influenza B viruses
Among the few influenza B viruses received (Table 1), many, in particular viruses from Italy, Madagascar and South Korea and from an outbreak in Israel in June 2004, were antigenically closely related to the vaccine strain B/Shandong/7/97, as well as the reference viruses, B/Tehran/80/02 and B/Brisbane/32/02, which are representative of the "Victoria HA/Yamagata NA" reassortants (Table 7). Their HA and NA sequences were also similar to those of earlier 2002-2003 isolates (Figures 9 and 11), but included some additional amino acid changes, indicative of recent genetic drift, K80E and A217S in HA and K125N and K373E in NA of viruses represented by e.g. B/Gwangju/275/03, and V90I in HA and D35G, L74I and N199S in NA of viruses represented by e.g. B/Israel/12/04 (Figures 10 and 12, Table 8).

Most of the sporadic isolates, from other European countries, were antigenically closely related to B/Shanghai/361/02 (Yamagata lineage) and the vaccine strain for 2004-2005, B/Jiangsu/10/03 (Table 7). The HA sequences of most of these viruses fell within the B/Harbin/7/94 ("Harbin") sublineage, but separated into two relatively homogenous phylogenetic groups (Figure 9). One, represented by B/Jiangsu/10/03, was characterized by 3 changes, K129N, D233A and G256R, relative to the HA of B/Shanghai/361/02; the other, represented by B/Oslo/71/04, had the characteristic amino acid change V252M (Figure 10, Table 8). Their NA sequences also fell into two corresponding phylogenetic groups (Figure 11). Those of viruses within the B/Oslo/71/04 group fell within the Harbin sublineage and possessed the amino acid changes T49I and S244P relative to the NAs of earlier isolates, such as B/Yaroslavl/11/02 (Figure 12, Table 8). They differed in some 10 residues from the NA sequences of B/Jiangsu/10/03 group viruses, which like the NA of B/Shanghai/361/02 fell within the Sichuan sublineage and were close to the NA sequence of B/Sichuan/379/99, but distinguished by two common changes P42T and D342N.

A H5N1 viruses

Collaborative studies showed that the human H5N1 viruses isolated in Vietnam and Thailand were similar in all 8 genes to viruses isolated from infected poultry in these and neighbouring countries during 2004. All virus genes were of avian origin and there was no evidence of reassortment with other human virus subtypes. With the exception of the NA gene, 7 genes were similar to those of H5N1 viruses isolated from human cases in Hong Kong in February 2003. The HAs like those of the H5N1 human viruses isolated in 1997 and 2003, have evolved from that of A/goose/Guangdong/1/96. They were however distinguishable antigenically from the 1997 and 2003 isolates in HI tests using post-infection ferret antisera. Amino acid changes relative to the HAs of the two 2003 isolates include A86V, N120S, A156T (which introduces a glycosylation site), R189K, K212R, N223S and A263T in HA1. The M genes of the 2004 human isolates, like the 2003 isolates, encoded asparagine at residue 31 in the M2 protein. Tests of drug susceptibility confirmed that these viruses were resistant to amantadine and rimantadine, but retained susceptibility to the anti-NA drugs, oseltamivir and zanamivir.

Swine Influenza viruses

Of 7 influenza A viruses isolated from pigs in France, 4 were of the H1N1 subtype and 3 were H1N2. Two of the H1N1 viruses were antigenically closely related to earlier isolates, such as A/swine/IV/1455/99 (Table 9). Reduced reactivity of the other two isolates in HI tests with the panel of reference antisera, together with changes in HA sequence, indicate the emergence of a new H1N1 antigenic variant. The H1N2 viruses were antigenically similar to A/swine/Côtes d'Armor/790/97 (Table 9) and similar in HA sequence to H1N2 viruses isolated during the previous year.

Reagents

Influenza viruses and antisera supplied to National Influenza Centres and other Collaborating Centres are listed in Table 10.

Acknowledgements

We specifically acknowledge the collaborative efforts of the staff of the many National Influenza Centres who submitted viruses for characterization. Conclusions regarding HA and NA sequences are based on sequence data shared among the four WHO Collaborating Centres and data received from the Influenza Centres in Denmark, France, Germany, Hong Kong, Ireland, Israel, Italy, Norway, Portugal, South Africa, Sweden and the U.K. (indicated by suffixes).

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Table 1: Human influenza isolates characterized August 2003 - July 2004

Country	H1N1 A/New Caledonia 20/99	H1N2 A/Egypt 96/02	H3N2 A/Panama 2007/99	A/Fujian 411/02	B B/Shanghai 361/02	B/HK 335/01
Algeria				3		
Australia				4		
Austria				13		
Belgium				23		
Brazil	1					
China				111		
Czech Republic				4		
Denmark		1		47		
Finland				6		
France	8	3	3	76	6	4
Germany				21		
Greece				10		
Hong Kong SAR China	3			31	3	
Iceland	10	23		10		
Ireland				3		1
Israel				44	2	3
Italy	1			113	1	8
Japan	1			2	1	
Latvia				18		
Madagascar				7		4
Malaysia				1		
Morocco	8			12		
Netherlands			1	7		
New Caledonia				1		
New Zealand				1		
Norway	2	11		25	5	
Portugal				14		
Romania				28		
Russia			2	26		1
Senegal	8	12		14		
Serbia and Montenegro				12		
Slovakia				12		
Slovenia				5		
South Africa				6		1
South Korea						19
Spain				116		
Sweden		4		49	1	
Switzerland				31	1	
Thailand					2	
Turkey			2			1
United Kingdom	2			7	2	
United States	2			4		
Zambia			2		1	
Total = 1094	**46**	**54**	**10**	**917**	**25**	**42**
	(4%)	**(5%)**	**927 (85%)**		**67 (6%)**	

Table 2. Human influenza viruses isolated August 2003 - July 2004

	H1N1	**H1N2**	**H3N2**		**B**	
Month of isolation	**A/New Caledonia 20/99**	**A/Egypt 96/02**	**A/Panama 2007/99**	**A/Fujian 411/02**	**B/Shanghai 361/02**	**B/HK 335/01**
2003						
August	1	2		22		
September		7		14		
October	11	18		60	3	
November	3	13		259	1	2
December	4	7		205		18
2004						
January	4			124	2	1
February		1	1	101	1	2
March	2		2	18	6	6
April	1			62	6	5
May	3			14	1	
June	3			8	4	4
July	1			25	1	
Total = 1059	**33**	**48**	**3**	**912**	**25**	**38**
	(3%)	**(4%)**	**915 (87%)**		**63 (6%)**	

Table 3. Antigenic analyses of influenza A H1N1 and H1N2 viruses

		Haemagglutination inhibition titre[1]				
		Post infection ferret sera				
Viruses	**Isolation Date**	**A/Beij 262/95**	**A/NC 20/99**	**A/Eg 96/02**	**A/Chile 8885/02**	**A/Hung 2/03**
A/Beijing/262/95		**640**	160	320	320	320
A/New Caledonia/20/99		80	**320**	640	640	1280
A/Egypt/96/02		40	160	**640**	320	640
A/Chile/8885/02		40	160	320	**640**	640
A/Hungary/2/03		40	80	320	640	**1280**
H1N1						
A/Reunion/1409/03	unk	80	160	640	640	640
A/Dakar/68/03	11.8.03	160	320	640	640	1280
A/Iceland/61/03	12.10.03	40	160	640	320	—
A/Paris/650/03	Nov-03	80	160	640	320	640
A/Poitiers/2168/03	18.11.03	80	160	640	160	320
A/Lyon GI/346/03	13.12.03	40	160	640	640	320
A/Morocco/275/04	24.12.03	80	320	640	320	1280
A/Morocco/434/04	28.1.04	40	160	320	320	320
A/Paris/2284/04	Apr-04	80	320	640	320	1280
A/England/40/04	20.5.04	80	320	320	320	640
A/Hong Kong/2765/04	29.6.04	<	80	160	160	160
A/Lyon/CHU/28824/04	8.7.04	80	320	640	640	320
H1N2						
A/Dakar/88/03	30.10.03	160	320	640	320	640
A/Oslo/149/03	3.11.03	160	160	640	320	1280
A/Umea/3/03	26.11.03	80	160	640	160	640
A/Iceland/123/03	8.12.03	80	160	640	320	320
A/Oslo/628/03	16.12.03	160	320	640	320	640
A/Denmark/86/03	18.12.03	80	320	640	160	320
A/Umea/1/04	6.2.04	80	160	320	160	640

1. < : <40

Table 4. Amino acid differences characteristic of HA and NA sequences of H1N1 and H1N2 viruses

Subtype	Representative strain	Amino acid changes[1] HA	NA
H1N1	A/England/40/04	V165A W251R Y252F (V314A)	V48I E332K N450D
	A/Hong Kong/2765/04	T82K Y94H R145K V165A R208K T266N	E214G R222Q V234M D382N
H1N2	A/Dakar/88/04	G152E V165A V174I A189T A214T	M24T[2] E41G E199K K431N S450T (G248E)
	A/Iceland/123/03	K73R V165A V174I G185R R188M A189T A214T Q282K	M24T[2] V30I E199K K308R K431N (G248E)

1. Relative to A/New Caledonia/20/99 (H1N1)
2. Relative to A/Singapore/15/01 (H3N2)

Table 5. Antigenic analyses of influenza A H3N2 viruses

		Haemagglutination inhibition titre[1]					
		Post infection ferret sera					
Viruses	**Isolation Date**	**A/Pan 2007/99**	**A/Egypt 130/02**	**A/Fuj 411/02**	**A/Wy 3/03**	**A/UK 1861/03**	**A/Chch 28/03**
A/Panama/2007/99		**2560**	1280	80	320	160	160
A/Egypt/130/02		1280	**2560**	80	160	160	160
A/Fujian/411/02		80	80	**640**	1280	640	1280
A/Wyoming/3/03		1280	640	2560	**5120**	1280	5120
A/UK/1861/03		80	80	40	640	**640**	320
A/Christchurch/28/03		80	160	640	1280	320	**1280**
A/Paris/637/03	Nov-03	640	1280	<	160	40	—
A/Samara/30/04	3.2.04	2560	2560	160	640	160	320
A/Turkey/405/04	3.3.04	1280	640	40	160	80	160
A/Denmark/17/03	1.9.03	160	160	320	1280	320	—
A/Ireland/10108/03	16.9.03	320	160	640	2560	320	—
A/Israel/21/03	9.10.03	160	160	320	640	—	1280
A/Morocco/50/04	18.10.03	160	160	320	1280	—	1280
A/Belgium/G24/04	23.10.03	160	320	320	1280	640	—
A/England/420/03	30.10.03	320	160	640	1280	640	—
A/Geneva/5361/03	7.11.03	160	160	320	640	320	—
A/Lisbon/17/03	16.11.03	160	160	640	1280	320	—
A/Netherlands/327/03	19.11.03	160	160	1280	2560	320	—
A/Barcelona/37/03	24.11.03	160	40	640	1280	320	—
A/Finland/324/03	28.11.03	320	160	2560	2560	640	—
A/Paris/861/04	Dec-03	160	160	320	2560	—	2560
A/Khaborovsk/34/03	1.12.03	160	160	1280	2560	640	1280
A/Prague/316/03	1.12.03	320	320	1280	5120	640	2560
A/Bucharest/194/03	2.12.03	80	80	160	640	320	1280
A/Lyon/2209/03	2.12.03	160	160	640	1280	640	1280
A/Oslo/487/03	4.12.03	80	80	320	1280	320	1280
A/Sachsen/366/03	11.12.03	320	320	1280	2560	640	5120
A/Algeria/303/03	14.12.03	40	40	80	320	160	—
A/Belgrade/7103/03	15.12.03	160	320	640	640	320	—
A/Stockholm/1/04	17.12.03	80	160	320	1280	320	1280
A/Slovenia/1410/03	18.12.03	160	160	320	640	320	—
A/Austria/134419/03	29.12.03	160	160	320	640	320	640
A/Madrid/G1418/03	6.1.03	160	320	640	1280	640	—
A/Wellington/1/2004	26.1.04	160	160	—	2560	—	2560
A/Latvia/1297/04	28.1.04	80	<	320	640	160	640
A/Hong Kong/439/04	24.2.04	80	160	160	640	160	640
A/Trieste/19/04	16.2.04	160	160	320	1280	320	1280
A/Iceland/6/04	18.2.04	80	160	160	640	—	1280
A/Madagascar/71824/04	18.2.04	80	40	160	2560	—	640
A/Morocco/467/04	17.2.04	80	160	640	5120	—	1280
A/Parma/61/04	Mar-04	80	160	160	640	320	1280
A/Slovakia/608/03	3.3.04	160	160	320	2560	—	1280
A/Shantou/575/04	14.4.04	160	160	—	640	—	2560
A/Dakar/15/04	9.6.04	160	160	320	1280	—	1280
A/Johannesburg/21/04	28.6.04	160	160	—	2560	—	1280
A/Shantou/1219/04	10.8.04	80	80	—	160	320	320

1. <,<40

Table 6. Amino acid changes characteristic of H3N2 sequence variants

Variant group	Representative strain	Amino Acid changes[1]	
		HA	NA
1	A/Wellington/1/04	Y159F S189N S227P	N93D E199K Q432E
	A/Shantou/1219/04	Y159F S189N K145N V226I	E199K I215V K431N
2	A/Shantou/575/04	N126D A138S K145N	S18A F23L I30V Y40H F42C V143G K172R E199K V216G T265I I307V S332F N385K D399E Q432E L437W
	A/Iceland/6/04	S45I G50E N126D N144D	as above T19A

1. Relative to A/Fujian/411/02.

Table 7. Antigenic analyses of influenza B viruses

Viruses	Isolation date	Haemagglutination inhibition titre[1]						
			Post infection ferret sera					
		B/Shan[2] 7/97	**B/Shan 7/97**	**B/Te 80/02**	**B/Bris 32/02**	**B/Sich 379/99**	**B/Shai 361/02**	**B/Jiang 10/03**
B/Shandong/7/97		**5120**	**160**	160	160	<	<	<
B/Tehran/80/02		2560	160	**320**	160	<	<	<
B/Brisbane/32/02		2560	160	80	**160**	<	<	<
B/Sichuan/379/99		<	<	<	<	**160**	160	<
B/Shanghai/361/02		<	<	<	<	320	**320**	40
B/Jiangsu/10/03		40	<	<	<	80	320	**320**
B/Geneva/6159/03	27.11.03	<	<	<	<	160	320	160
B/Netherlands/87/04	Jan-04	<	<	<	<	160	160	160
B/Oslo/71/04	13.1.04	<	<	<	<	160	320	320
B/Lyon/GI/74/04	2.2.04	<	<	<	<	160	160	80
B/Israel/4/04	17.2.04	<	<	<	<	80	160	160
B/Milan/66/04	Mar-04	<	<	<	<	160	320	320
B/Stockholm/1/04	12.3.04	<	<	<	<	160	320	320
B/England/23/04	17.3.04	<	<	<	<	160	320	320
B/Paris/2282/04	Apr-04	<	<	<	<	160	320	320
B/Oslo/586/04	27.5.04	<	<	<	<	160	320	160
B/Shantou/1212/04	4.8.04	<	<	<	<	160	320	640
B/Ireland/13064/03	17.12.03	5120	160	80	80	<	<	<
B/Cheju/303/03	18.12.03	1280	80	40	40	<	<	<
B/Cheonbuk/318/03	27.12.03	320	40	<	40	<	<	<
B/Madagascar/71537/04	12.1.04	2560	80	40	160	<	<	<
B/Parma/8/04	Apr-04	2560	160	160	160	<	<	<
B/Caserta/1/04	Apr-04	2560	80	160	160	<	<	<
B/Israel/13/04	7.6.04	1280	80	80	80	<	<	<
B/Madagascar/72134/04	28.6.04	2560	80	40	320	<	<	<
B/Johannesburg/22/04	30.6.04	2560	160	160	160	<	<	<

[1] <, <40

[2] hyperimmune sheep serum

Table 8. Amino acid changes characteristic of recent B sequence variants

HA Lineage	Representative strain	Amino acid changes	
		HA	NA
Harbin(1)	B/Jiangsu/10/03	H40Y[1]	
		K129N	
		D233A	
		G256R	
Harbin(2)	B/Oslo/71/04	H40Y[1]	T42Q[1]
		V252M	T49I
			K186R
			N219K
			N235D
			S244P
			D329N
			N342D
			D392E
			E436T
Victoria	B/Gwangju/275/03	K80E[2]	K125N[3]
		I121T	I248V
		A217S	K373E
	B/Israel/12/04	V90I[2]	D35G[3]
		I121T	L74I
			N199S
			I248V

1. Relative to B/Shanghai/361/02. 2. Relative to B/Shandong/7/97. 3. Relative to B/Tehran/80/02.

Table 9 . Antigenic analyses of swine H1N1 and H1N2 viruses

		Haemagglutination inhibition titre[1]								
				post infection ferret sera						
VIRUSES	**SUBTYPE**	**A/Braz[2] 11/78**	**Sw/Fin[2] 2899/82**	**A/Braz 11/78**	**Sw/Fin 2899/82**	**Sw/It 1513-1/98**	**Sw/CA 1482/99**	**Sw/IV 1455/99**	**Sw/Scot 410440/94**	**Sw/CA 790/97**
A/Brazil/11/78	H1N1	**5120**	640	**1280**	80	40	40	40	160	80
Sw/Fin/2899/82	H1N1	40	**5120**	<	**1280**	320	640	320	<	<
Sw/Italy/1513-1/98	H1N1	160	5120	<	1280	**640**	640	320	<	<
Sw/CA/1482/99	H1N1	160	5120	<	1280	640	**1280**	640	<	<
Sw/IV/1455/99	H1N1	<	1280	<	160	40	80	**640**	<	<
Sw/Scotland/410440/94	H1N2	5120	160	640	<	40	<	<	**5120**	1280
Sw/CA/790/97	H1N2	5120	40	320	<	<	<	<	160	**2560**
Sw/Morbihan/1345/03	H1N2	5120	40	80	<	<	<	<	160	1280
Sw/CA/1374/03	H1N2	5120	40	80	<	<	<	<	160	1280
Sw/CA/1589/04	H1N2	5120	80	160	<	<	<	<	320	1280
Sw/CA/1404/03	H1N1	40	5120	<	160	80	80	<	<	<
Sw/CA/1405/03	H1N1	40	5120	<	1280	160	1280	640	<	<
Sw/CA/103/04	H1N1	<	5120	<	640	160	320	640	<	<
Sw/Fin/492/04	H1N1	40	5120	<	80	40	40	40	<	<

1. <,<40; 2. hyperimmune rabbit serum

Table 10. Viruses and sera despatched August 2003 - July 2004

Country	Viruses		Sera	
	A	**B**	**A**	**B**
Australia	4	1		
Austria	7	4	7	4
Belgium	3	2	3	2
Bulgaria	4	1	3	1
Croatia	1	1	1	1
Czech Republic	1	1	1	1
Denmark	2	1	3	3
Finland	1	1	1	1
France	5	5	7	4
Germany	4	4	4	5
Greece	6	3	6	3
Hong Kong	1			
Hungary	3	3	2	1
Ireland	3	2	3	2
Israel	1		1	
Italy	7	5	7	5
Japan	4	1		
Latvia	3	2	3	2
Luxembourg	1	1	1	1
Netherlands	2	2	2	2
Norway	3	3	3	2
Poland	3	2	3	2
Portugal	3	2	3	2
Romania	4	4	3	2
Slovakia	3	2	3	2
Slovenia	3	2	3	2
South Korea	2	2	5	2
Spain	3	2	3	2
Sweden	3	2	3	2
Switzerland	4	3	6	2
United Kingdom	3	2	1	1
United States	8	1		
Total = 322	**105**	**67**	**91**	**59**
	172		**150**	

Figure 1. Phylogenetic comparison of nucleotide sequences encoding H1 haemagglutinins

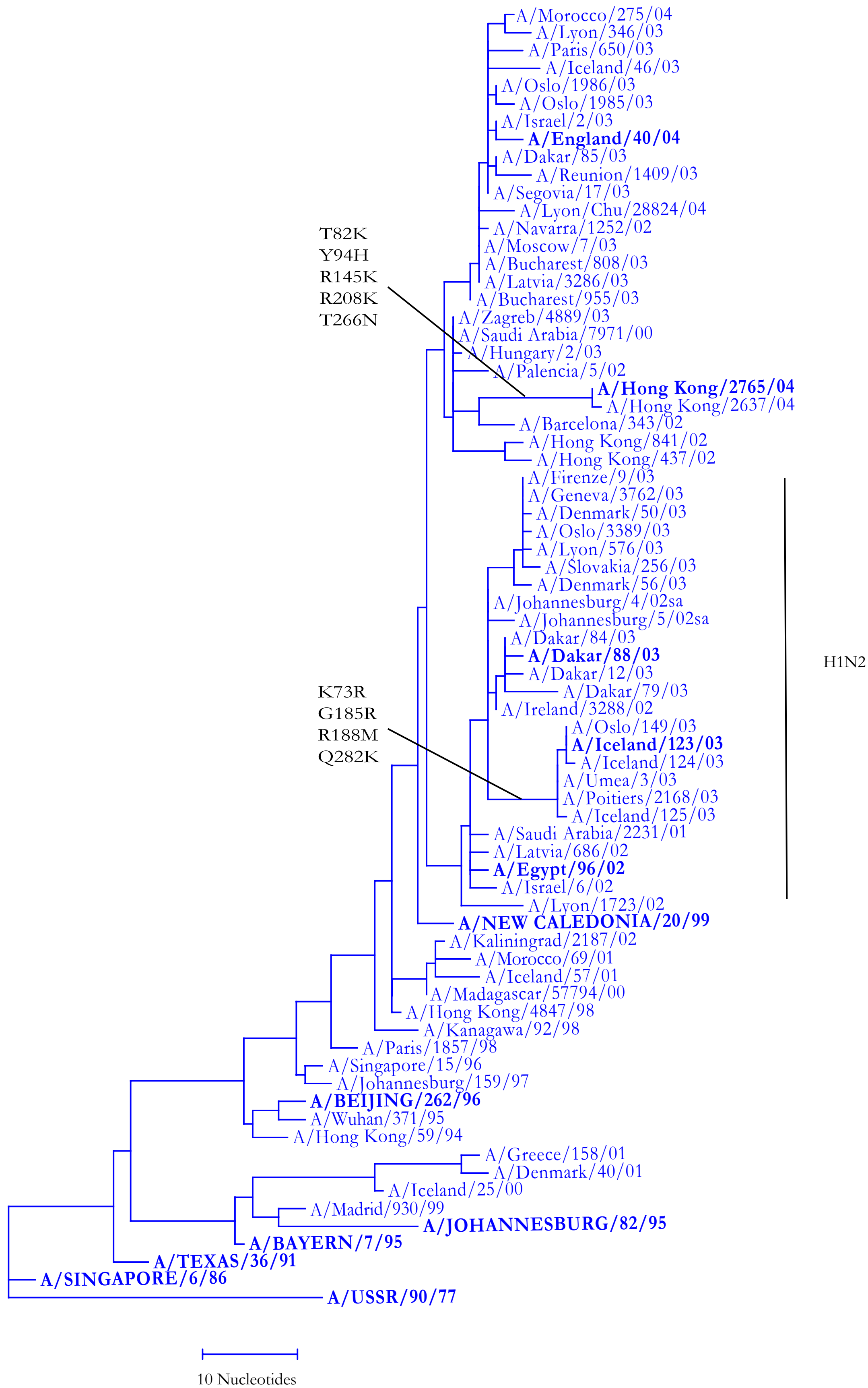

Figure 2. HA sequences (amino acids 1-300) of H1N1 and H1N2 viruses

	1									100
A/Beijing/262/95	__________	__________	__________	__________	__________	__________	________s_	__________	__________	__________
A/New Caledonia/20/99	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Zagreb/4889/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hungary/2/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Segovia/17/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Dakar/85/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Bucharest/955/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Bucharest/808/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Reunion/1409/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Lyon/Chu/28824/04	__________	__________	__________	__________	__________	__________	__________	_p________	__________	__________
A/Paris/650/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Lyon/346/03	__________	__________	__________	_____n____	__________	__________	__________	___k______	__________	__________
A/Morocco/275/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Israel/2/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/England/40/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/841/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/437/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Barcelona/343/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/2765/04	__________	__________	__________	__________	__________	__________	__________	__________	_k________	___h______
A/Hong Kong/2637/04	__________	__________	__________	__________	__________	__________	__________	__________	_k________	___h______
A/Egypt/96/02	__________	__________	__________	__________	__________	__________	__________	__________	___s______	__________
A/Firenze/9/03	__________	__________	__________	__________	__________	__________	__________	__________	________a_	__________
A/Geneva/37623/03	__________	__________	__________	__________	__________	__________	__________	__________	________a_	__________
A/Denmark/50/03	__________	__________	__________	__________	__________	__________	__________	__________	________a_	__________
A/Dakar/88/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Dakar/84/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Iceland/125/03	__________	__________	__________	__________	__________	__________	__________	__r_______	__________	__________
A/Umea/3/03	__________	__________	__________	__________	__________	__________	__________	__r_______	__________	__________
A/Poitiers/2168/03	__________	__________	__________	__________	__________	__________	__________	__r_______	__________	__________
A/Oslo/149/03	__________	__________	__________	__________	__________	__________	__________	__r_______	__________	__________
A/Iceland/123/03	__________	__________	__________	__________	__________	__________	__________	__r_______	__________	__________
Consensus	DTICIGYHAN	NSTDTVDTVL	EKNVTVTHSV	NLLEDSHNGK	LCLLKGIAPL	QLGNCSVAGW	ILGNPECELL	ISKESWSYIV	ETPNPENGTC	YPGYFADYEE

Figure 2 (Continued)

	101									200
A/Beijing/262/95	__________	__________	__________	_t________	__________	_e________	_n__v_____	__________	_s___v___i	__________
A/New Caledonia/20/99	__________	__________	__________	__________	__________	__________	____v_____	__________	_____d____	__________
A/Zagreb/4889/03	__________	__________	__________	__________	__________	__________	__________	__________	_____d____	__________
A/Hungary/2/03	__________	__________	__________	__________	__________	__________	__________	__________	_____d____	__________
A/Segovia/17/03	__________	__________	__________	__________	__________	__________	__________	__________	_____d____	__________
A/Dakar/85/03	__________	__________	__________	__________	__________	__________	__________	__________	_____d____	__________
A/Bucharest/955/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Bucharest/808/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Reunion/1409/03	_______i__	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Lyon/Chu/28824/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Paris/650/03	__________	__________	__________	__________	__________	__________	__________	__________	_____d____	__________
A/Lyon/346/03	__________	__________	__________	__________	__________	__________	__________	__________	_____d____	__________
A/Morocco/275/04	__________	__________	__________	__________	__________	__________	__________	__________	_____d____	__________
A/Israel/2/03	__________	__________	__________	__________	__________	__________	__________	__________	_____d____	__________
A/England/40/04	__________	__________	__________	__________	__________	__________	__________	__________	_____d____	__________
A/Hong Kong/841/02	__________	__________	__________	_________r	__________	__________	__________	__________	_____d____	__________
A/Hong Kong/437/02	__________	__________	__________	__t_______	__________	__________	__________	__________	_____n____	__________
A/Barcelona/343/02	__________	____k_____	__________	__________	____k_____	__________	__________	__________	____rv____	__________
A/Hong Kong/2765/04	__________	__________	__________	__________	____k_____	__________	__________	__________	__________	__________
A/Hong Kong/2637/04	__________	__________	__________	__________	____k_____	__________	__________	__________	__________	__________
A/Egypt/96/02	__________	__________	_______i__	__________	__________	__________	__________	__________	________t_	__________
A/Firenze/9/03	__________	__________	__________	__________	__________	__________	__________	___i______	________t_	__________
A/Geneva/37623/03	__________	__________	__________	__________	__________	__________	__________	___i______	________t_	__________
A/Denmark/50/03	__________	__________	__________	__________	__________	__________	__________	___i______	________t_	__________
A/Dakar/88/03	__________	__________	__________	__________	__________	_e________	__________	___i______	________t_	__________
A/Dakar/84/03	__________	__________	__________	__________	__________	_e________	__________	___i______	_____d__t_	__________
A/Iceland/125/03	__________	__________	__________	______r___	__________	__________	__________	___i______	____r__mt_	__________
A/Umea/3/03	__________	__________	__________	__________	__________	__________	__________	___i______	____r__mt_	__________
A/Poitiers/2168/03	__________	__________	__________	__________	__________	__________	__________	___i______	____r__mt_	__________
A/Oslo/149/03	__________	__________	__________	__________	__________	__________	__________	___i______	____rd_mt_	__________
A/Iceland/123/03	__________	__________	__________	__________	__________	__________	__________	___i______	____rd_mt_	__________
Consensus	LREQLSSVSS	FERFEIFPKE	SSWPNHTVTG	VSASCSHNGK	SSFYRNLLWL	TGKNGLYPNL	SKSYANNKEK	EVLVLWGVHH	PPNIGNQRAL	YHTENAYVSV

Figure 2 (continued)

	201									300
A/Beijing/262/95	__________	__________	g_________	__________	__________	w_________	__________	_n________	__________	__________
A/New Caledonia/20/99	__________	__________	__________	__________	__________	w_________	__________	__________	__________	__________
A/Zagreb/4889/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hungary/2/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Segovia/17/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Dakar/85/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Bucharest/955/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Bucharest/808/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Reunion/1409/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Lyon/Chu/28824/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Paris/650/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Lyon/346/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Morocco/275/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Israel/2/03	__________	__________	__________	__________	__________	_f________	__________	__________	__________	__________
A/England/40/04	__________	__________	__________	__________	__________	_f________	__________	__________	__________	__________
A/Hong Kong/841/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/437/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Barcelona/343/02	__________	__________	__________	__________	__________	__________	__________	i_________	__________	__________
A/Hong Kong/2765/04	_______k__	__________	__________	__________	__________	__________	_____n____	__________	__________	__________
A/Hong Kong/2637/04	_______k__	__________	__________	__________	__________	__________	_____n____	______e___	__________	__________
A/Egypt/96/02	__________	___t______	__________	__________	__________	w_________	__________	__________	__________	__________
A/Firenze/9/03	__________	___t______	__________	__________	__________	w_________	__________	__________	__________	__________
A/Geneva/37623/03	__________	___t______	__________	__________	__________	w_________	__________	__________	__________	__________
A/Denmark/50/03	__________	___t______	__________	__________	__________	w_________	__________	__________	__________	__________
A/Dakar/88/03	__________	___t______	__________	__________	__________	w_________	__________	__________	__________	___i______
A/Dakar/84/03	__________	___t______	__________	__________	__________	w_________	__________	__________	__________	__________
A/Iceland/125/03	__________	___t______	__________	__________	__________	w_________	__________	__________	_k________	__________
A/Umea/3/03	__________	___t______	__________	__________	__________	w_________	__________	__________	_k________	__________
A/Poitiers/2168/03	__________	___t______	__________	__________	__________	w_________	__________	__________	_k________	__________
A/Oslo/149/03	__________	___t______	__________	__________	__________	w_________	__________	__________	_k________	__________
A/Iceland/123/03	__________	___t______	__________	__________	__________	w_________	__________	__________	_k________	__________
Consensus	VSSHYSRRFT	PEIAKRPKVR	DQEGRINYYW	TLLEPGDTII	FEANGNLIAP	RYAFALSRGF	GSGIITSNAP	MDECDAKCQT	PQGAINSSLP	FQNVHPVTIG

Figure 3. Phylogenetic comparison of nucleotide sequences encoding N1 neuraminidases

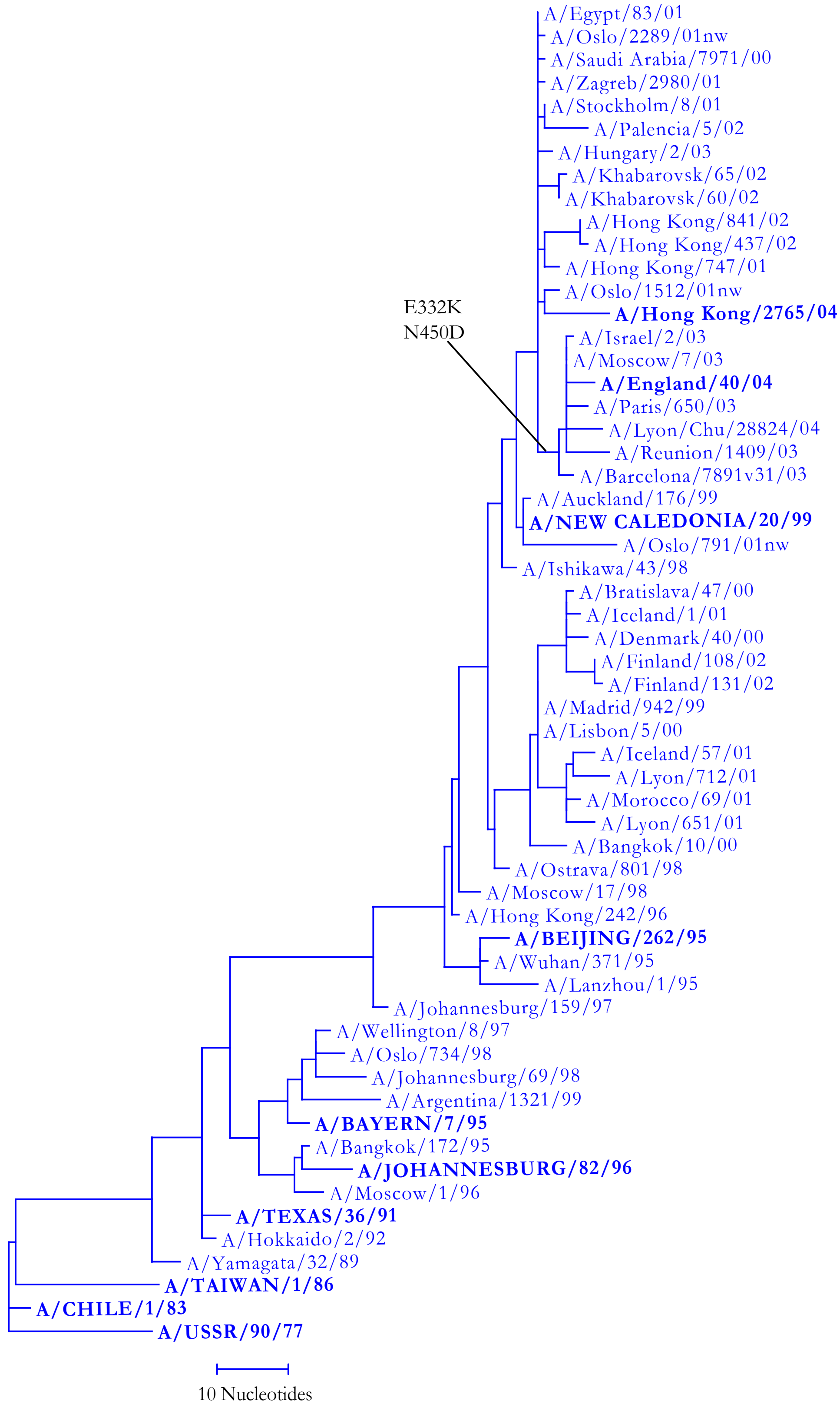

Figure 4. NA sequences of H1N1 viruses

```
                          1                                                                                                   100
A/Beijing/262/95          __________ _____v____ __________ __________ __________ __________ __________ __________ __________ __________
A/New Caledonia/20/99     __________ __________ __________ __________ _______v__ __________ __________ __________ __________ __________
A/Oslo/1512/01nw          ~~~_______ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Zagreb/2980/01          ~~~~~~____ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Stockholm/8/01          ~~~~~~~~~~ ~~________ __________ __________ __________ __________ __________ __________ __________ __________
A/Khabarovsk/65/02        ~~________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Hong Kong/841/02        __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Hong Kong/437/02        ~~~~______ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Palencia/5/02           ~~~~~~~___ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Hungary/2/03            ~_________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Hong Kong/2765/04       __________ __________ __________ __________ _______v__ __________ __________ __________ __________ __________
A/Israel/l2/03            __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Moscow/7/03             __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Barcelona/7891v31/03    ~_________ __________ __________ __________ _______v__ __________ __________ __________ __________ __________
A/Reunion/1409/03         __________ __________ __________ __________ __________ _k________ __________ __________ __________ __________
A/Paris/650/03            ~~~~~_____ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/England/40/04           ~~~~~_____ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Lyon/Chu/28824/04       ~~~~~~~~~~ ~~~~~_____ __________ __________ __________ __________ __________ __________ __________ __________
               Consensus  MNPNQKIITI GSISIAIGII SLMLQIGNII SIWASHSIQT GSQNHTGICN QRIITYENST WVNHTYVNIN NTNVVAGKDK TSVTLAGNSS LCSISGWAIY

                          101                                                                                                 200
A/Beijing/262/95          __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/New Caledonia/20/99     __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Oslo/1512/01nw          __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Zagreb/2980/01          __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Stockolm/8/01           __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Khabarovsk/65/02        __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Hong Kong/841/02        __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Hong Kong/437/02        __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Palencia/5/02           __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Hungary/2/03            __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Hong Kong/2765/04       __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Israel/l2/03            __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Moscow/7/03             __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Barcelona/7891v31/03    __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Reunion/1409/03         __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Paris/650/03            __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/England/40/04           __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Lyon/Chu/28824/04       __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
               Consensus  TKDNSIRIGS KGDVFVIREP FISCSHLECR TFFLTQGALL NDKHSNGTVK DRSPYRALMS CPLGEAPSPY NSKFESVAWS ASACHDGMGW LTIGISGPDN
```

Figure 4 (Continued)

	201									300
A/Beijing/262/95	__________	__________	__________	__________	__________	__________	__________	___s______	__________	__________
A/New Caledonia/20/99	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Oslo/1512/01nw	__________	__________	__________	__________	__________	________k_	__________	__________	__________	__________
A/Zagreb/2980/01	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Stockholm/8/01	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Khabarovsk/65/02	__________	__________	__________	__________	__________	__________	_________s	__________	__________	__________
A/Hong Kong/841/02	__________	__________	__________	__________	__________	__________	______v___	__________	__________	__________
A/Hong Kong/437/02	__________	__________	__________	__________	__________	__________	______v___	__________	__________	__________
A/Palencia/5/02	__________	___g______	__________	__________	__________	__________	__________	__________	__________	__________
A/Hungary/2/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/2765/04	__________	___g______	_q________	___m______	__________	__________	__________	__________	__________	__________
A/Israel/l2/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Moscow/7/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Barcelona/7891v31/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Reunion/1409/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Paris/650/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/England/40/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Lyon/Chu/28824/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
Consensus	GAVAVLKYNG	IITETIKSWK	KRILRTQESE	CVCVNGSCFT	IMTDGPSNGA	ASYKIFKIEK	GKVTKSIELN	APNFHYEECS	CYPDTGTVMC	VCRDNWHGSN

	301									400
A/Beijing/262/95	__________	__________	__________	__________	__________	_r________	__________	__________	__________	_____m____
A/New Caledonia/20/99	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Oslo/1512/01nw	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Zagreb/2980/01	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Stockholm/8/01	__________	__________	__________	__________	__________	__________	_____k____	__________	__________	__________
A/Khabarovsk/65/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/841/02	______d___	__________	__________	__________	__________	__________	__________	__________	_______t__	___g______
A/Hong Kong/437/02	______d___	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Palencia/5/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hungary/2/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/2765/04	__________	__________	__________	__________	__________	__________	__________	__________	_n________	__________
A/Israel/l2/03	__________	__________	_k________	__________	__________	__________	__________	__________	__________	__________
A/Moscow/7/03	__________	__________	__________	_k________	__________	__________	____d_____	__________	__________	__________
A/Barcelona/7891v31/03	__________	__________	__________	_k________	__________	__________	__________	__________	__________	__________
A/Paris/650/03	__________	__________	__________	_k________	__________	__________	__________	__________	_n________	__________
A/England/40/04	__________	__________	__________	_k________	__________	__________	__________	__________	__________	__________
A/Lyon/Chu/28824/04	__________	__________	__________	_k________	__________	__________	__________	__________	__________	__i_______
Consensus	RPWVSFNQNL	DYQIGYICSG	VFGDNPRPKD	GEGSCNPVTV	DGADGVKGFS	YKYGNGVWIG	RTKSNRLRKG	FEMIWDPNGW	TDTDSDFSVK	QDVVAITDWS

Figure 4 (Continued)

```
                                401                                                                 470
A/Beijing/262/95                __________ __________ _________r __________ __________ __________ __________
A/New Caledonia/20/99           __________ __________ __________ __________ __________ __________ __________
A/Oslo/1512/01nw                __________ __________ __________ __________ __________ _________~ ~~~~~~~~~~
A/Zagreb/2980/01                __________ __________ __________ __________ __________ __________ _______~~~
A/Stockholm/8/01                __________ __________ __________ __________ __________ __________ __________
A/Khabarovsk/65/02              __________ __________ __________ __________ __________ __________ __________
A/Hong Kong/841/02              __________ __________ __________ __________ __________ _a________ __________
A/Hong Kong/437/02              __________ __________ __________ __________ __________ _a________ __________
A/Palencia/5/02                 __________ __________ __________ __________ __________ __________ __________
A/Hungary/2/03                  __________ __________ __________ __________ __________ __________ __________
A/Hong Kong/2765/04             __________ __________ __________ __________ __________ __________ __________
A/Israel/l2/03                  __________ __________ __________ __________ _________d __________ __________
A/Moscow/7/03                   __________ __________ __________ __________ _________d __________ __________
A/Barcelona/7891v31/03          __________ __________ __________ __________ _________d __________ __________
A/Reunion/1409/03               __________ __________ __________ __________ _________d __________ __________
A/Paris/650/03                  __________ __________ __________ __________ _________d __________ __________
A/England/40/04                 __________ __________ __________ __________ _________d __________ __________
A/Lyon/Chu/28824/04             __________ __________ __________ __________ _________d __________ __________
                     Consensus  GYSGSFVQHP ELTGLDCIRP CFWVELVRGL PRENTTIWTS GSSISFCGVN SDTANWSWPD GAELPFTIDK
```

Figure 5. Phylogenetic comparison of nucleotide sequences encoding H3 haemagglutinins

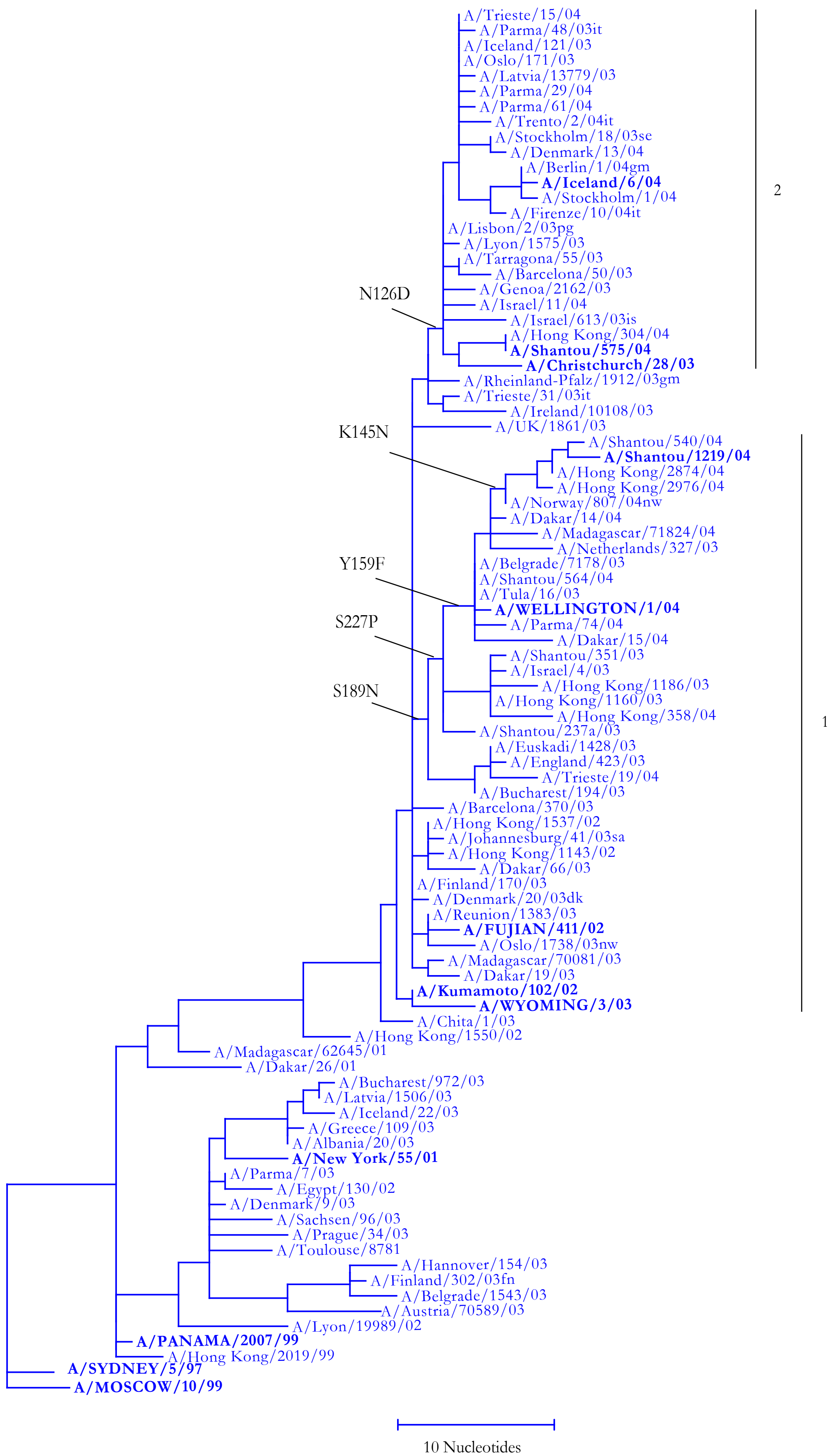

Figure 6. HA sequence (amino acid 1-300) of H3N2 viruses

	1									100
A/Sydney/5/97	__i_______	__________	____l_____	__________	_________r	______r___	__________	____h_____	__e_______	__________
A/Panama/2007/99	__________	__________	s___l_____	__________	_________r	__________	__________	____h_____	__e_______	__________
A/New York/55/01	__________	__________	____l_____	__________	________sr	__________	__________	____h_____	__e_______	__________
A/Egypt/130/02	__________	__________	____l_____	__________	_________r	__________	__________	____h_____	__e_______	__________
A/Hong Kong/1550/02	__________	__________	____l_____	__________	__________	__________	__________	____h_____	__________	_____s____
A/Chita/1/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Fujian/411/02	~~~~~~~~~~	~~~~~~~~~~	~~~~~~~~~~	~~________	__________	__________	__________	__________	__________	__________
A/Wyoming/3/03	~~~~~~~~~~	~~~~~~~~~~	~~~~~~~~~~	~~~~______	__________	__________	__________	__________	__________	__________
A/Hong Kong/1186/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Israel/4/03	~~~~~~~~~~	~~~~~~~~~~	~~~~~~~~~~	~~~~______	__________	__________	__________	__________	__________	__________
A/England/423/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	_r________
A/Trieste/19/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Shantou/237a/03	__________	__________	__________	__________	_________e	__________	__________	__________	__________	__________
A/Netherlands/327/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Tula/16/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Parma/74/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Wellington/1/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Madagascar/71824/04	__________	__________	__________	__________	_________e	__________	__________	__________	__________	__________
A/Dakar/14/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/2976/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Shantou/1219/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Norway/807/04nw	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/UK/1861/03	~~~~~~~~~~	~~~~~~~~~~	~~~~~~~~~~	~~~~~~~___	__________	__________	__________	__________	__________	__________
A/Christchurch/28/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Israel/11/04	__________	__________	__________	__________	__________	__n_______	__________	__________	__________	__________
A/Hong Kong/304/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Shantou/575/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Parma/29/04	__________	__________	__________	__________	____i_____	__________	__________	__________	__________	__________
A/Iceland/6/04	__________	__________	__________	__________	____i____e	__________	__________	__________	__________	__________
A/Stockholm/1/04	__________	__________	____l_____	__________	____i____e	__________	__________	__________	__________	__________
A/Berlin/1/04gm	~~~~~~~~~~	~~~~~~~~~~	~~~~~~~~~~	~~klk_____	____i____e	__________	__________	__________	__________	__________
Consensus	QKLPGNDNST	ATLCLGHHAV	PNGTIVKTIT	NDQIEVTNAT	ELVQSSSTGG	ICDSPHQILD	GENCTLIDAL	LGDPQCDGFQ	NKKWDLFVER	SKAYSNCYPY

Figure 6 (continued)

	101									200
A/Sydney/5/97	__________	__________	__________	a_____y___	_s_i______	____hq____	__________	_d________	_____s____	_t_i______
A/Panama/2007/99	__________	__________	__________	a_________	__________	____hq____	__________	__________	__l__s____	__________
A/New York/55/01	_____v____	__________	__________	a_________	___d______	____hq____	__________	__________	__________	___i______
A/Egypt/130/02	_____v____	__________	__________	a_________	___d______	____hq____	__s_______	__________	__________	__________
A/Hong Kong/1550/02	__________	__________	__________	a_________	__________	____hq____	__________	__________	_____v____	___i______
A/Chita/1/03	__________	__________	__________	__________	__________	_____q____	__________	__________	_____v____	__________
A/Fujian/411/02	__________	__________	__________	__________	__________	__________	__________	__________	__l_______	__________
A/Wyoming/3/03	__________	__________	_______a__	__________	__________	__________	__________	__________	_____v____	__________
A/Hong Kong/1186/03	__________	__________	___g______	__________	__________	__________	__________	__________	__________	__n_______
A/Israel/4/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__n_______
A/England/423/03	__________	__________	__________	__________	_k________	__________	__________	__________	________n_	__________
A/Trieste/19/04	__________	__________	__________	__________	_k________	______s___	__________	__________	________n_	__________
A/Shantou/237a/03	__________	__________	__________	__________	__________	__________	__________	__________	________n_	_______e__
A/Netherlands/327/03	__________	__________	__________	__________	__________	________f_	__________	__________	________n_	__________
A/Tula/16/03	__________	__________	__________	__________	__________	________f_	__________	__________	________n_	__________
A/Parma/74/04	__________	__________	__________	__________	__________	________f_	__________	__________	________n_	__________
A/Wellington/1/04	__________	__________	__________	__________	__________	________f_	__________	__________	________n_	__________
A/Madagascar/71824/04	__________	__________	__________	__________	__________	________f_	__________	__________	________n_	__________
A/Dakar/14/04	__________	__________	__________	__________	__________	________f_	__________	__________	__________	__________
A/Hong Kong/2976/04	__________	__________	__________	__________	____n_____	________f_	__________	__________	________n_	__________
A/Shantou/1219/04	__________	__________	__________	_______s__	____d_____	________f_	__________	__________	________n_	__________
A/Norway/807/04nw	__________	__________	__________	__________	____n_____	________f_	__________	__________	________n_	__________
A/UK/1861/03	__________	__________	__________	__________	__________	______p___	__________	__________	__l_______	_____t____
A/Christchurch/28/03	____h_____	__________	_____d____	_______s__	__________	__________	__________	__________	_____d___v	__________
A/Israel/11/04	__________	__________	_____d____	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/304/04	__________	__________	_____d____	_______s__	____n_____	__________	__________	__________	__________	__________
A/Shantou/575/04	__________	__________	_____d____	_______s__	____n_____	__________	__________	__________	__________	__________
A/Parma/29/04	__________	__________	_____d____	__________	__________	__________	__________	__________	__________	__________
A/Iceland/6/04	__________	__________	_____d____	__________	___d______	__________	__________	__________	__________	__________
A/Stockholm/1/04	__________	__________	_____d____	__________	___d______	__________	__________	__________	__________	__________
A/Berlin/1/04gm	__________	__________	_____d____	__________	___d______	__________	__________	__________	__________	__________
Consensus	DVPDYASLRS	LVASSGTLEF	NNESFNWTGV	TQNGTSSACK	RRSNKSFFSR	LNWLTHLKYK	YPALNVTMPN	NEKFDKLYIW	GVHHPGTDSD	QISLYAQASG

Figure 6 (continued)

	201									300
A/Sydney/5/97	_v________	__________	_w__gi____	__________	__________	__________	__________	__________	__________	__________
A/Panama/2007/99	_v________	__________	_w__g_____	__________	__________	__________	__________	__________	__________	__________
A/New York/55/01	_v________	__________	_w__g___g_	__________	__________	__________	__________	__________	__________	__________
A/Egypt/130/02	_v________	_________g	_w__g_____	__________	__________	__________	__r_______	__________	__________	________k_
A/Hong Kong/1550/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Chita/1/03	__________	__________	______p___	__________	_______i__	__________	__________	__________	__________	__________
A/Fujian/411/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Wyoming/3/03	__________	________y_	_____i____	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/1186/03	__________	__________	______p___	__________	__________	__________	__________	__________	__________	__________
A/Israel/4/03	__________	__________	______p___	__________	__________	__________	__________	__________	__________	__________
A/England/423/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Trieste/19/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Shantou/237a/03	__________	__________	______p___	__________	__________	__________	__________	__________	__________	__________
A/Netherlands/327/03	__________	__________	_____i____	__________	__________	__________	__________	__________	__________	__________
A/Tula/16/03	__________	__________	______p___	__________	__________	__________	__________	__________	__________	__________
A/Parma/74/04	__________	__________	______p___	__________	__________	__________	__________	__________	__________	__________
A/Wellington/1/04	__________	__________	______p___	__________	__________	__________	__________	__________	__________	__________
A/Madagascar/71824/04	__________	__________	_____ip___	__________	__________	__________	________k_	__________	__________	__________
A/Dakar/14/04	__________	__________	_____ip___	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/2976/04	__________	__________	_____ip___	__________	__________	__________	__________	__________	__________	__________
A/Shantou/1219/04	__________	__________	_____i____	__________	__________	__________	__________	__________	__________	__________
A/Norway/807/04nw	__________	__________	_____ip___	__________	__________	__________	__________	__________	__________	__________
A/UK/1861/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Christchurch/28/03	__________	__________	_____i____	__________	__________	__________	__________	__________	__________	__________
A/Israel/11/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Hong Kong/304/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Shantou/575/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Parma/29/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Iceland/6/04	__________	___t______	__________	__________	__________	__________	__________	__________	__________	__________
A/Stockholm/1/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
A/Berlin/1/04gm	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
Consensus	RITVSTKRSQ	QTVIPNIGSR	PRVRDVSSRI	SIYWTIVKPG	DILLINSTGN	LIAPRGYFKI	RSGKSSIMRS	DAPIGKCNSE	CITPNGSIPN	DKPFQNVNRI

Figure 7. Phylogenetic comparison of nucleotide sequences encoding N2 neuraminidases

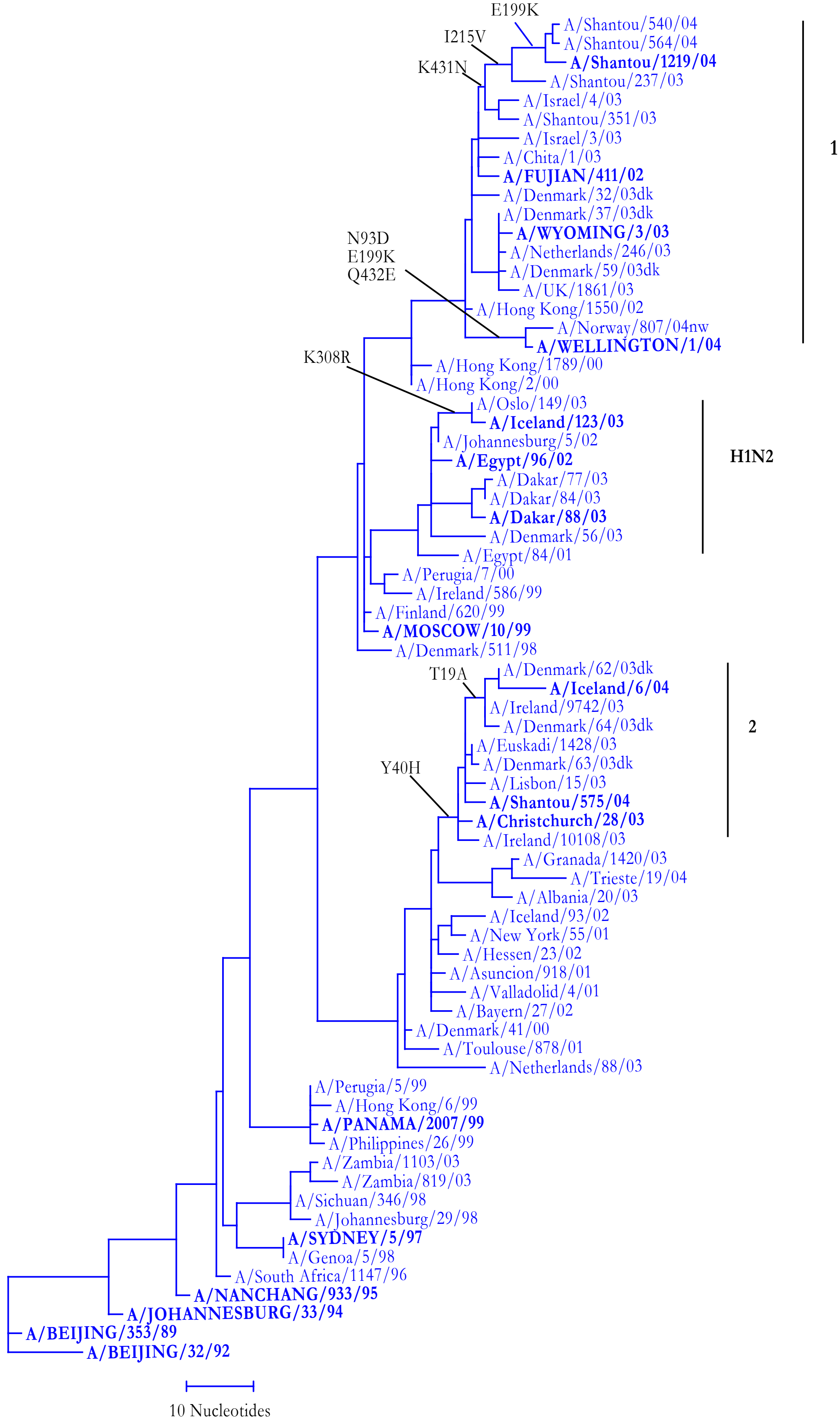

Figure 8. NA sequences of H3N2 and H1N2 viruses

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                        1                                                                                                              100
A/Sydney/5/97           __________ __________ __________ __________ __s_______ __________ __________ __________ __________ __k_______
A/Panama/2007/99        __________ __________ __________ __________ __________ __________ __________ __________ __________ __k_______
A/Moscow/10/99          __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Egypt/96/02           __________ __________ ___t______ __________ __________ __________ __________ __________ __________ __________
A/Dakar/88/03           __________ __________ ___t______ __________ g_________ __________ __________ __________ __________ __________
A/Oslo/149/03           __________ __________ ___t_____i __________ __________ __________ __________ __________ __________ __________
A/Iceland/123/03        __________ __________ ___t_____i __________ __________ __________ __________ __________ __________ __________
A/New York/55/01        __________ __________ __________ __________ __________ _____a____ __________ __________ __________ __________
A/Granada/1420/03       __________ __________ ______t___ __________ __________ __________ __________ __________ __________ __________
A/Trieste/19/04         __________ __________ ______t___ __________ __________ __________ __________ __________ __________ __________
A/Christchurch/28/03    __________ __________ __________ _________h __________ __________ __________ __________ _v________ __________
A/Lisbon/15/03          __________ __________ __________ _________h __________ __________ __________ __________ __________ __________
A/Shantou/575/04        __________ __________ __________ _________h __________ __________ __________ __________ __________ __________
A/Denmark/63/03dk       __________ __________ __________ _________h __________ __________ __________ __________ __________ __________
A/Ireland/9742/03       ~~~_______ ________a_ __________ _________h __________ __________ __________ __________ __________ __________
A/Iceland/6/04          __________ ________a_ __________ _________h __________ __________ __________ __________ __________ __________
A/Fujian/411/02         __________ _______s__ __f______i __________ _f________ __________ __________ __________ __________ __________
A/Wyoming/3/03          __________ _______s__ __f______i __________ _f________ __________ __________ __________ __________ __________
A/Israel/4/03           __________ _______s__ __f______i __________ _f________ __________ __________ __________ __________ __________
A/Shantou/351/03        __________ _______s__ __f______i _________h _f________ __________ __________ __________ __________ __________
A/Shantou/564/04        __________ _______s__ __f______i __________ _f________ __________ __________ __________ __________ __________
A/Shantou/1219/04       ~~________ _______s__ __f______i __________ _f________ __________ __________ __________ __________ __________
A/Wellington/1/04       __________ _______s__ __f______i __________ _f________ __________ __________ __________ __________ __d_______
A/Norway/807/04nw       __________ _______s__ __f______i __________ _f________ __________ __________ __________ __________ __d_______
             Consensus  MNPNQKIITI GSVSLTIATI CFLMQIAILV TTVTLHFKQY ECNSPPNNQV MLCEPTIIER NITEIVYLTN TTIEKEICPK LAEYRNWSKP QCNITGFAPF

                        101                                                                                                            200
A/Sydney/5/97           __________ __________ __________ __________ __r_______ __________ __________ __________ __________ ______h_e_
A/Panama/2007/99        __________ __________ __________ __________ __r_______ __________ __________ __________ __________ ______h_e_
A/Moscow/10/99          __________ __________ __________ __________ __________ __________ __________ __________ __________ ________e_
A/Egypt/96/02           __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Dakar/88/03           __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Oslo/149/03           __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Iceland/123/03        __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/New York/55/01        __________ __________ __________ __________ __________ __________ __________ _r________ __________ ________e_
A/Granada/1420/03       __________ __________ __________ __________ __________ __________ __________ _r________ __________ __________
A/Trieste/19/04         __________ __________ __________ __________ __________ __________ __________ _r________ __________ __________
A/Christchurch/28/03    __________ __________ __________ __________ __________ __________ __________ _r________ __________ __________
A/Lisbon/15/03          __________ __________ __________ __________ __________ __________ __________ _r________ __________ __________
A/Shantou/575/04        __________ __________ __________ __________ __________ g_________ __________ _r________ __________ __________
A/Denmark/63/03dk       __________ __________ __________ __________ __________ __________ __________ _r________ __________ __________
A/Ireland/9742/03       __________ __________ __________ __________ __________ __________ __________ _r________ __________ __________
A/Iceland/6/04          __________ __________ __________ __________ __________ g_________ __________ _r________ __________ __________
A/Fujian/411/02         __________ __________ __________ __________ __v_______ __________ __________ __________ __________ ________e_
A/Wyoming/3/03          __________ __________ __________ __________ __v_______ __________ __________ __________ __________ ________e_
A/Israel/4/03           __________ __________ __________ __________ __v_______ g_________ __________ __________ __________ ________e_
A/Shantou/351/03        __________ __________ __________ __________ __v_______ __________ __________ __________ __________ ________e_
A/Shantou/564/04        __________ __________ __________ __________ __v_______ __________ __________ __________ __________ __________
A/Shantou/1219/04       __________ __________ __________ __________ __v_______ __________ __________ __________ __________ __________
A/Wellington/1/04       __________ __________ __________ __________ __v_______ __________ __________ __________ __________ __________
A/Norway/807/04nw       __________ __________ __________ __________ __v_______ __________ __________ __________ __________ __________
             Consensus  SKDNSIRLSA GGDIWVTREP YVSCDPDKCY QFALGQGTTL NNGHSNDTVH DRTPYRTLLM NELGVPFHLG TKQVCIAWSS SSCHDGKAWL HVCVTGDDKN
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Figure 8. (Continued)

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                      201                                                                                                           300
A/Sydney/5/97         _______d__ __________ __________ __________ ________r_ __________ ____i_p___ __________ __________ __________
A/Panama/2007/99      _______d__ __________ __________ __________ ________r_ __________ ______k___ __________ __________ __________
A/Moscow/10/99}       __________ __________ __________ __________ __________ __________ ______p___ __________ __________ __________
A/Egypt/96/02         __________ __________ __________ __________ __________ __________ ______l___ __________ __________ __________
A/Dakar/88/03         __________ __________ __________ __________ _______e__ __________ ____i_l___ __________ __________ __________
A/Oslo/149/03         __________ __________ __________ __________ __________ __________ ______l___ __________ __________ __________
A/Iceland/123/03      __________ __________ __________ __________ _______e__ __________ ______l___ __________ __________ __________
A/New York/55/01      __________ __________ __________ __________ __________ __________ ____i_____ __________ __________ __________
A/Granada/1420/03     __________ __________ __________ __________ __________ __________ __i_i_____ __________ __________ __________
A/Trieste/19/04       __________ __________ __________ __________ __________ __________ __i_i_____ __________ __________ __________
A/Christchurch/28/03  __________ __________ __________ __________ __________ __________ ____i_____ __________ __________ __________
A/Lisbon/15/03        __________ __________ __________ __________ __________ __________ ____i_____ __________ __________ __________
A/Shantou/575/04      __________ __________ __________ __________ __________ __________ ____i_____ __________ __________ __________
A/Denmark/63/03dk     __________ __________ __________ __________ __________ __________ ____i_____ __________ __________ __________
A/Ireland/9742/03     __________ __________ __________ __________ __________ __________ ____i_____ __________ __________ __________
A/Iceland/6/04        __________ __________ __________ __________ __________ __________ ____i_____ __________ __________ __________
A/Fujian/411/02       __________ _____v____ __________ __________ __________ __________ __________ __________ __________ __________
A/Wyoming/3/03        __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Israel/4/03         __________ _____v____ __________ __________ __________ __________ __________ __________ __________ __________
A/Shantou/351/03      __________ _____v____ __________ __________ __________ __________ __________ __________ __________ __________
A/Shantou/564/04      __________ ____vv____ __________ __________ __________ __________ __________ __________ __________ __________
A/Shantou/1219/04     __________ ____vv____ __________ __________ __________ __________ __________ __________ __________ __________
A/Wellington/1/04     __________ _____v____ __________ __________ __________ __________ __________ __________ __________ __________
A/Norway/807/04nw     __________ _____a____ e_________ __________ __________ __________ __________ __________ __________ __________
           Consensus  ATASFIYNGR LVDSIGSWSK KILRTQESEC VCINGTCTVV MTDGSASGKA DTKILFIEEG KIVHTSTLSG SAQHVEECSC YPRYPGVRCV CRDNWKGSNR

                      301                                                                                                           400
A/Sydney/5/97         __________ __________ __________ ________n_ __________ __________ _________f __________ __________ __________
A/Panama/2007/99      __________ __________ __________ __________ __________ __________ _________s __________ __________ ________e_
A/Moscow/10/99}       __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Egypt/96/02         __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Dakar/88/03         __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Oslo/149/03         _______r__ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/Iceland/123/03      _______r__ __________ __________ __________ __________ __________ __________ __________ __________ __________
A/New York/55/01      __________ __________ __________ _f________ __________ __________ __________ __________ __________ ________e_
A/Granada/1420/03     __________ __________ __________ __________ __________ __________ __________ __________ __________ ________e_
A/Trieste/19/04       __________ __________ __________ __________ __________ __________ __________ __________ __________ ________e_
A/Christchurch/28/03  __________ __________ __________ _f________ __________ __________ __________ __________ __________ ________e_
A/Lisbon/15/03        __________ __________ __________ _f________ __________ __________ __________ __________ __________ ________e_
A/Shantou/575/04      __________ __________ __________ _f________ __________ __________ __________ __________ __________ ________e_
A/Denmark/63/03dk     __________ __________ __________ _f________ __________ __________ __________ __________ __________ ________e_
A/Ireland/9742/03     __________ __________ __________ _f________ __________ __________ __________ __________ __________ ________e_
A/Iceland/6/04        __________ __________ __________ _f________ __________ __________ __________ __________ __________ ________e_
A/Fujian/411/02       ______i___ __________ __________ __________ __________ __________ __________ __________ ____n_____ __________
A/Wyoming/3/03        ______i___ __________ __________ __________ __________ __________ __________ __________ ____n_____ __________
A/Israel/4/03         ______i___ __________ __________ __________ __________ __________ __________ __________ ____n_____ __________
A/Shantou/351/03      ______i___ __________ __________ __________ __________ __________ __________ __________ ____n_____ __________
A/Shantou/564/04      ______i___ __________ __________ __________ __________ __________ __________ __________ ____n_____ __________
A/Shantou/1219/04     ______i___ __________ __________ __________ __________ __________ __________ __________ ____n_____ __________
A/Wellington/1/04     ______i___ __________ __________ __________ __________ __________ __________ __________ ____n_____ __________
A/Norway/807/04nw     ______i___ __________ __________ __________ __________ __________ __________ __________ ____n_____ __________
           Consensus  PIVDINVKDY SIVSSYVCSG LVGDTPRKND SSSSSHCLDP NNEEGGHGVK GWAFDDGNDV WMGRTISEKL RSGYETFKVI EGWSKPNSKL QINRQVIVDR
```

Figure 8. (Continued)

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                         401                                                                  469
A/Sydney/5/97            __________ __________ __________ ______w___ __________ __________ _________
A/Panama/2007/99         __m_______ __________ __________ ______w___ __________ __________ _________
A/Moscow/10/99}          __________ __________ __________ __________ __________ __________ _________
A/Egypt/96/02            __________ __________ __________ n_________ __________ __________ _________
A/Dakar/88/03            __________ __________ __________ n_________ _________t __________ _________
A/Oslo/149/03            __________ __________ __________ n_________ __________ __________ _________
A/Iceland/123/03         __________ __________ __________ n_________ __________ __________ _________
A/New York/55/01         __________ __________ __________ _e____w___ __________ __________ _________
A/Granada/1420/03        __________ __________ __________ _e____w___ __________ __________ _________
A/Trieste/19/04          s_________ __________ __________ _e____w___ __________ __________ ____~~~~~
A/Christchurch/28/03     __________ __________ __________ _e____w___ __________ __________ _r~~~~~~~
A/Lisbon/15/03           __________ __________ __________ _e____w___ __________ __________ _________
A/Shantou/575/04         __________ __________ __________ _e____w___ __________ __________ _________
A/Denmark/63/03dk        __________ __________ __________ _e____w___ __________ __________ _________
A/Ireland/9742/03        __________ __________ __________ _e____w___ __________ __________ _______~~
A/Iceland/6/04           __________ __________ __________ _e____w___ __________ __________ ______~~~
A/Fujian/411/02          __________ __________ __________ __________ __________ __________ ________~
A/Wyoming/3/03           __________ __________ __________ __________ __________ __________ ______~~~
A/Israel/4/03            d_________ __________ __________ n_________ __________ __________ ______~~~
A/Shantou/351/03         d_________ __________ __________ n_________ __________ __________ _____~~~~
A/Shantou/564/04         __________ __________ __________ n_________ __________ __________ _________
A/Shantou/1219/04        __________ __________ __________ n_________ __________ __________ ________~
A/Wellington/1/04        __________ __________ __________ _e__k_____ __________ __________ _________
A/Norway/807/04nw        __________ __________ __________ _e________ __________ __________ _________
            Consensus    GNRSGYSGIF SVEGKSCINR CFYVELIRGR KQETEVLWTS NSIVVFCGTS GTYGTGSWPD GADINLMPI
```

Figure 9. Phylogenetic comparison of nucleotide sequences encoding B haemagglutinins

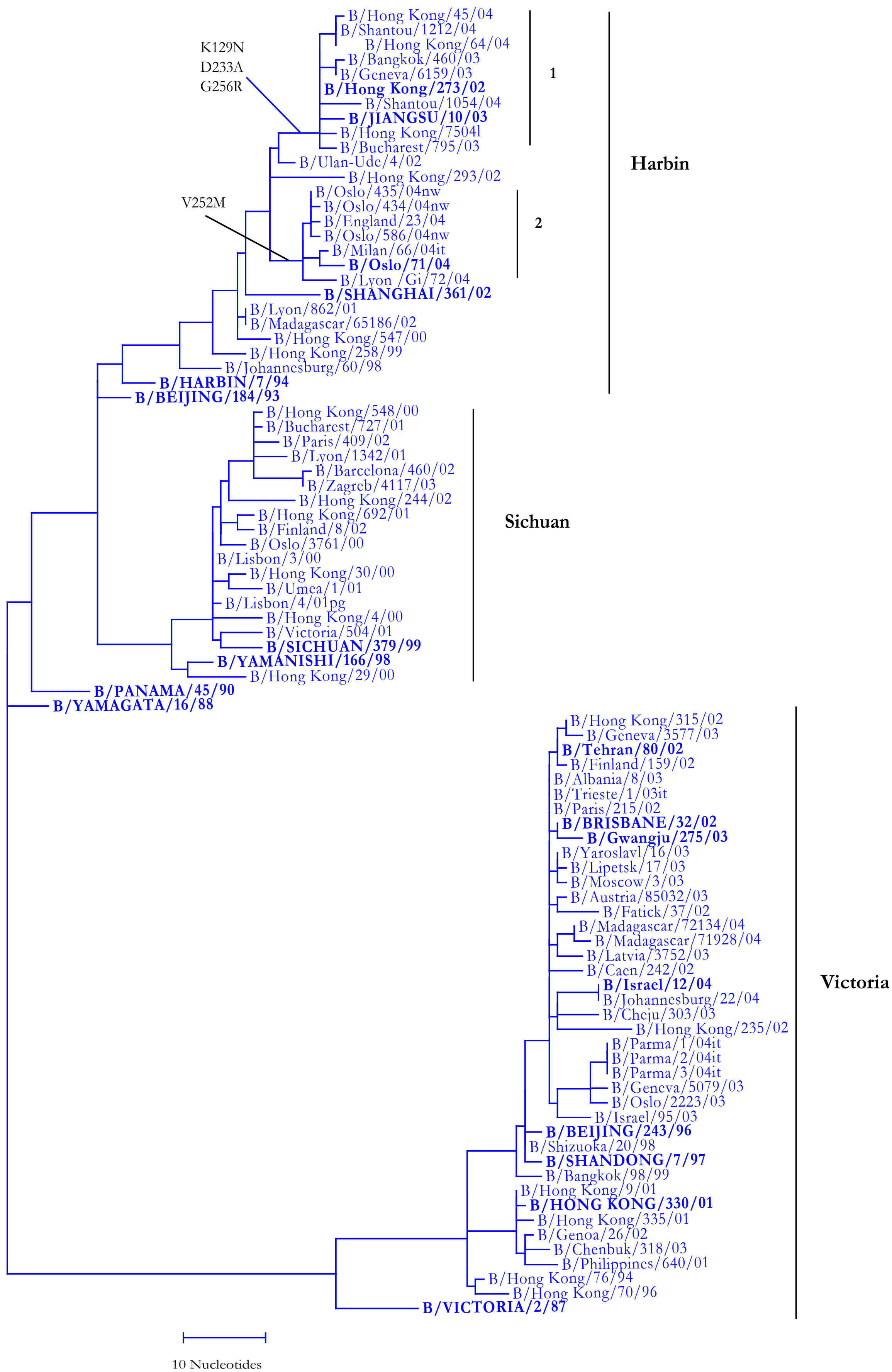

Figure 10. HA sequences (amino acids 1-300) of B viruses

	1									100
B/Beijing/184/93	__________	__________	__________	__________	_g________	_____n____	__________	__________	__________	__________
B/Harbin/7/94	__________	__________	__________	__________	__________	_____n____	__________	__________	__________	__________
B/Shanghai/361/02	__________	__________	__________	______i___	_______r__	__________	__________	__________	__________	__________
B/Hong Kong/273/02	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Bucharest/795/03	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Jiangsu/10/03	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Bangkok/460/03	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Geneva/6159/03	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Shantou/1054/04	__________	__________	__________	_________y	__________	__________	__________	__________	__________	__________
B/Hong Kong/45/04	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Shantou/1212/04	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Oslo/71/04	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Milan/66/04it	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/England/23/04	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Lyon/GI/72/04	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Oslo/586/04nw	__________	__________	__________	_________y	_______r__	__________	__________	__________	__________	__________
B/Sichuan/379/99	__________	__________	________a_	__________	__________	_____t____	__________	____i_____	______ik__	__________
B/Paris/409/02	__________	__________	________a_	__________	__________	_____t_f__	__________	____i_____	__________	__________
B/Zagreb/4117/03	__________	__________	________a_	__________	__________	_____t_f__	__________	____i__p__	__________	__________
B/Shandong/7/97	__________	__________	__________	__________	__________	_____k____	__________	k_t_ni____	v_________	__________
B/Tehran/80/02	__________	__________	__________	__________	__________	_____k____	__________	k_t_ni____	v_________	__________
B/Geneva/3577/03	__________	__________	__________	__________	__________	_____k____	__________	k_t_ni___e	v_________	__________
B/Gwangju/275/03	__________	__________	__________	__________	__________	_____k____	__________	k_t_ni___e	v_________	__________
B/Cheju/303/03	__________	__________	__________	__________	__________	_____e____	__________	k_t_ni____	v_________	__________
B/Oslo/2223/03	__________	k_________	__________	__________	__________	_____k____	__________	k_t_ni____	v_________	______m___
B/Geneva/5079/03	~~~~~~~~~~	~~~~~~~~~~	~~~~~~~~~~	__________	__________	_____k____	__________	k_t_ni____	v_________	__________
B/Parma/1/04it	__________	__________	__________	__________	__________	_____k____	__________	k_t_ni____	v_________	__________
B/Madagascar/72134/04	__________	d_________	__________	__________	__________	_____k____	__________	k_t_ni____	v_________	__________
B/Israel/12/04	__________	__________	__________	__________	__________	_____k____	__________	k_m_ni____	v________i	__________
B/Johannesburg/22/04	__________	__________	__________	__________	__________	_____k____	__________	k_m_ni____	v________i	__________
B/Hong Kong/330/01	__________	__________	__________	__________	__________	_____k____	__________	k_t_ni____	v_________	__________
B/Geneva/26/02	~~~~~~~~~~	~~~~~~~~~~	~~~~______	__________	__________	_____k____	__________	k_t_ni____	v_________	__________
B/Chenbuk/318/03	__________	__________	__________	__________	__________	_____k____	__________	k_t_ni____	v_________	__________
Consensus	DRICTGITSS	NSPHVVKTAT	QGEVNVTGVI	PLTTTPTKSH	FANLKGTKTR	GKLCPDCLNC	TDLDVALGRP	MCVGTTPSAK	ASILHEVRPV	TSGCFPIMHD

Figure 10 (continued)

	101									200
B/Beijing/184/93	__________	__________	__________	__________	________r_	__________	_r._______	__________	__k_______	________i_
B/Harbin/7/94	__________	__________	__________	__________	________r_	__________	_r._d_____	_________v	__________	________a_
B/Shanghai/361/02	__________	__________	_____d____	l_________	__________	__________	__.____n__	__________	__________	______d___
B/Hong Kong/273/02	__________	__________	_____d__n_	__________	__________	__________	__.____n__	_________v	__________	__________
B/Bucharest/795/03	__________	__________	_____d__n_	__________	__________	__________	__.____n__	_________v	__________	______s___
B/Jiangsu/10/03	__________	__________	_____d__n_	__________	r_________	__________	__.____n__	_________v	__________	__________
B/Bangkok/460/03	__________	__________	_____d__n_	__________	__________	__________	__.n___n__	__________	__________	__________
B/Geneva/6159/03	__________	__________	_____d__n_	__________	__________	__________	__.____n__	__________	__________	__________
B/Shantou/1054/04	__________	__________	_____d__n_	__________	__________	__________	__.____n__	_________v	__________	__________
B/Hong Kong/45/04	__________	__________	_____d__n_	__________	__________	__________	__.____n__	_________v	__________	__________
B/Shantou/1212/04	__________	__________	_____d__n_	__________	__________	__________	__.____n__	_________v	__________	__________
B/Oslo/71/04	__________	__________	_____d____	__________	__________	__________	__.____n__	__________	__________	__________
B/Milan/66/04it	__________	__________	_____d____	__________	__________	__________	__.____n__	__________	__________	__________
B/England/23/04	__________	__________	_____d____	__________	__________	__________	__.____n__	__________	__________	__________
B/Lyon/GI/72/04	__________	__________	_____d____	__________	__________	__________	__.____n__	__________	__________	________i_
B/Oslo/586/04nw	__________	__________	_____d____	__________	__________	__________	__.____n__	__________	__________	__________
B/Sichuan/379/99	__________	_____k____	__________	__________	__________	__________	_r._______	________h_	__ke______	__________
B/Paris/409/02	__________	_____k____	_____d____	__________	__________	__________	_r._______	________h_	__ke______	__________
B/Zagreb/4117/03	__________	_____k____	_____d____	__________	_______n__	__________	_r._______	________h_	__ke______	__________
B/Shandong/7/97	__________	_____h____	ih________	_____ki___	_____v_ngn	__________	__________	_s__i_____	__________	_______e_k
B/Tehran/80/02	__________	_____h____	_h________	_____ki___	_____v_ngn	__________	__________	_s__i_____	__________	_______ei_
B/Geneva/3577/03	__________	_____h____	_h________	_____ki___	_____v_ngn	__________	__________	_s__i_____	__________	_______ei_
B/Gwangju/275/03	__________	_____h____	_h________	_____ki___	_____v_ngn	__________	__________	_s__i_____	__________	_______ea_
B/Cheju/303/03	__________	_____h____	_h________	_____ki___	_____v_ngn	__________	__________	_s__i_____	__________	_______e__
B/Oslo/2223/03	__________	_____h____	_h________	_____ki___	_____v_ngn	__________	__________	_s__i_____	__________	_______e__
B/Geneva/5079/03	__________	_____h____	_h________	_____ki___	_____v_ngn	__________	__________	____i_____	__________	_______e__
B/Parma/1/04it	__________	_____h____	_h________	_____ki___	_____v_ngn	__________	__________	_s__i_____	__________	_______e__
B/Madagascar/72134/04	__________	_____h____	_h________	_____ki___	_____i_ngn	__________	__________	_s__i_____	__________	_______e__
B/Israel/12/04	__________	_____h____	_h________	_____ki___	_____v_ngn	__________	__________	_s__i_____	__________	_______e__
B/Johannesburg/22/04	__________	_____h____	_h________	_____ki___	_____v_ngn	__________	__________	_s__i_____	__________	_______e__
B/Hong Kong/330/01	__________	_____r____	nh________	_____ki___	_____v_ngn	__________	___e______	_s__i_____	__________	______se__
B/Geneva/26/02	__________	_____r____	nh________	_____ki___	_____v_ngn	__________	___e______	_s__i_____	__________	_______e__
B/Chenbuk/318/03	__________	_____r____	nh______e_	_____ki___	_____i_ngn	__________	___e______	_s__i_____	__________	_______e__
Consensus	RTKIRQLPNL	LRGYENIRLS	TQNVINAEKA	PGGPYRLGTS	GSCPNATSKS	GFFATMAWAV	PKNDNNKTAT	NPLTVEVPYI	CTEGEDQITV	WGFHSDNKTQ

Figure 10 (continued)

	201									300
B/Beijing/184/93	__________	__________	__________	__d_______	__________	__________	______v___	__________	__________	__________
B/Harbin/7/94	__________	__________	__________	__d_______	__________	__________	______v___	__________	__________	__________
B/Shanghai/361/02	__________	__________	__________	__d_______	__________	__________	______v___	__________	__________	__________
B/Hong Kong/273/02	__________	__________	__________	__a_______	__________	_____r____	______v___	__________	__________	__________
B/Bucharest/795/03	__________	__________	__________	__a_______	__________	_____r____	______v___	__________	__________	__________
B/Jiangsu/10/03	__________	__________	__________	__a____e__	__________	_____r____	______v___	__________	__________	__________
B/Bangkok/460/03	__________	__________	__________	__a_______	__________	_____r____	______v___	__________	__________	__________
B/Geneva/6159/03	__________	__________	__________	__a_______	__________	_____r____	______v___	__________	__________	__________
B/Shantou/1054/04	__________	__________	__________	__a_______	__________	_____r____	______v___	__________	__________	__________
B/Hong Kong/45/04	__________	__________	a_________	__a_______	__________	_____r____	______v___	__________	__________	__________
B/Shantou/1212/04	__________	__________	__________	__a_______	__________	_____r____	______v___	__________	__________	__________
B/Oslo/71/04	__________	__________	__________	__d_______	__________	_m________	______v___	__________	__________	__________
B/Milan/66/04it	__________	__________	__________	__d_______	__________	_m________	______v___	__________	__________	__________
B/England/23/04	__________	__________	__________	__d_______	__________	_m________	______v___	__________	__________	__________
B/Lyon/GI/72/04	__________	__________	_a________	__d_______	__________	_m________	______v___	__________	__________	__________
B/Oslo/586/04nw	__________	__________	__________	__d_______	__________	_m________	______v___	__________	__________	__________
B/Sichuan/379/99	__________	_________i	__________	__d_______	__________	__________	__________	__________	__________	__________
B/Paris/409/02	__________	_________i	__________	__d_______	__________	__________	__________	__________	__________	__________
B/Zagreb/4117/03	__________	_________i	__________	__d_______	__________	__________	__________	__________	__________	__________
B/Shandong/7/97	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Tehran/80/02	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Geneva/3577/03	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Gwangju/275/03	_ak_____k_	______s___	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Cheju/303/03	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Oslo/2223/03	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Geneva/5079/03	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Parma/1/04it	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Madagascar/72134/04	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Israel/12/04	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Johannesburg/22/04	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Hong Kong/330/01	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Geneva/26/02	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
B/Chenbuk/318/03	_ak_____k_	__________	__________	__________	__________	____s_____	_t________	__________	__________	__________
Consensus	MKNLYGDSNP	QKFTSSANGV	TTHYVSQIGG	FPNQTEDGGL	PQSGRIVVDY	MVQKPGKTGT	IVYQRGILLP	QKVWCASGRS	KVIKGSLPLI	GEADCLHEKY

Figure 11. Phylogenetic comparison of nucleotide sequences encoding B neuraminidases

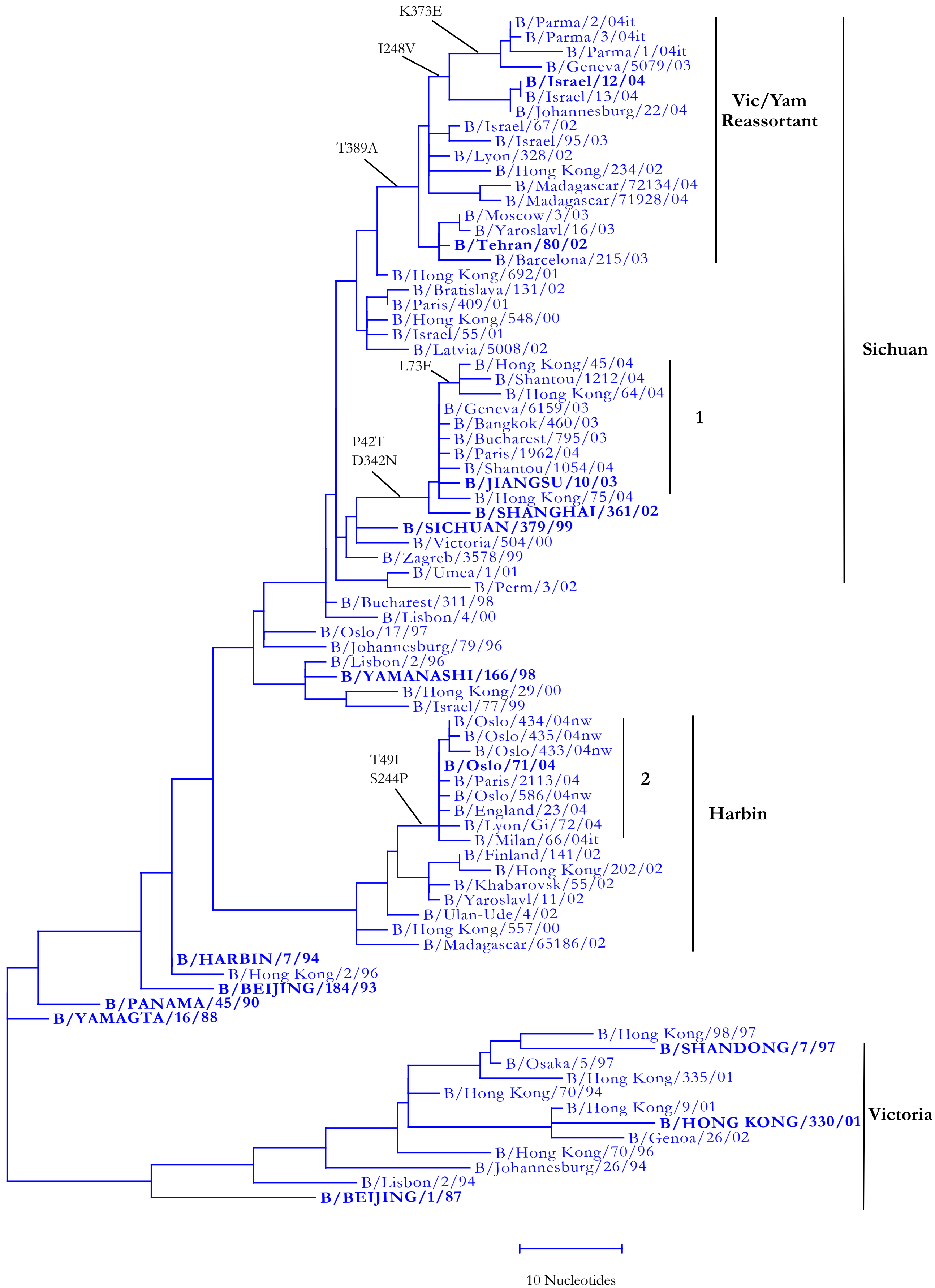

Figure 12. NA sequences of B viruses

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                         1                                                                                                          100
B/Harbin/7/94            __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Yaros1avl/1/02         __________ __________ __________ __________ _q________ __________ __________ __________ __________ __________
B/England/23/04          __________ __________ __________ __________ _q______i_ __________ __________ __________ __________ __________
B/Oslo/71/04             __________ ____l_____ __________ __________ _q______i_ __________ __________ __________ __________ __________
B/Lyon/GI/72/04          __________ __________ __________ __________ _q______i_ __________ __________ __________ __________ __________
B/Milan/66/04it          __________ __________ __________ __________ _q___i__i_ __________ ____h_____ __________ __________ __________
B/Paris/2113/04          __________ __________ __________ __________ _q______i_ __________ __________ __________ __________ __________
B/Oslo/586/04nw          __________ __________ __________ __________ _q______i_ __________ __________ __________ __________ __________
B/Sichuan/379/99         __________ __________ __________ __________ __________ __________ __________ __f_______ __________ __________
B/Shanghai/361/02        __________ __________ __________ __________ _t________ __________ __________ __________ __________ __________
B/Jiangsu/10/03          __________ __________ __________ __________ _t__t_____ __________ __________ __________ __________ __________
B/Bangkok/460/03         __________ __________ __c_______ __________ _t________ __________ ____h_____ __________ __________ __________
B/Bucharest/795/03       __________ __________ __________ __________ _t________ __________ __________ __________ __________ __________
B/Geneva/6159/03         __________ __________ __________ __________ _t________ __________ __________ __________ __________ __________
B/Paris/1962/04          __________ __________ __________ __________ _t________ __________ __m_______ __________ __________ __________
B/Hong Kong/64/04        __________ __________ __________ __________ _t________ __n_______ __________ __f_______ __________ __________
B/Shantou/1212/04        __________ __________ __________ __________ _t________ __________ __________ __f_______ __________ __________
B/Tehran/80/02           __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Moscow/3/03            __________ __________ __________ __________ __________ __________ ____h_____ __________ __________ __________
B/Madagascar/72134/04    __________ __________ l_________ __________ __________ __________ __________ __________ __________ __________
B/Geneva/5079/03         __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Parma/1/04it           __________ __________ __________ __________ __________ __________ __________ _______s__ __________ __________
B/Israel/12/04           ~~~~~~~___ __________ __________ ____g_____ __________ __________ __________ ____i_____ __________ __________
B/Johannesburg/22/04     __________ __________ __________ ____g_____ __________ __________ __________ ____i_____ __________ __________
            Consensus    MLPSTIQTLT LFLTSGGVLL SLYVSASLSY LLYSDILLKF SPTEITAPTM PLDCANASNV QAVNRSATKG VTLLLPEPEW TYPRLSCPGS TFQKALLISP

                         101                                                                                                        200
B/Harbin/7/94            ______r___ __________ __________ __________ __________ __________ __________ __________ _____r____ __________
B/Yaros1avl/1/02         __________ __________ __________ __________ __________ __________ __________ __________ _____r____ __________
B/England/23/04          __________ __________ __________ __________ __________ __________ __________ __________ _____r____ __________
B/Oslo/71/04             __________ __________ __________ __________ __________ __________ __________ __________ _____r____ __________
B/Lyon/GI/72/04          __________ __________ __________ __________ __________ __________ __________ __________ _____r____ __________
B/Milan/66/04it          __________ __________ __________ __________ __________ __________ __________ __________ _____r____ __________
B/Paris/2113/04          __________ __________ __________ __________ __________ __________ __________ __________ _____r____ __________
B/Oslo/586/04nw          __________ __________ __________ __________ __________ __________ __________ __________ _____r____ __________
B/Sichuan/379/99         __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Shanghai/361/02        __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Jiangsu/10/03          __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Bangkok/460/03         __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Bucharest/795/03       __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Geneva/6159/03         __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Paris/1962/04          __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Hong Kong/64/04        __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Shantou/1212/04        __________ __________ __________ __________ __________ __________ __________ __________ __________ __________
B/Tehran/80/02           __________ __________ __________ __________ _______g__ __________ __________ __________ __________ _______n__
B/Moscow/3/03            __________ __________ __________ __________ _______g__ __________ __________ __________ __________ _______nd_
B/Madagascar/72134/04    __________ __________ __________ __________ _______g__ __________ __________ __________ __________ _______n__
B/Geneva/5079/03         __________ __________ __________ __________ _______g__ __________ __________ __________ __________ _______nd_
B/Parma/1/04it           __________ __________ __________ __________ _______g__ _________n __________ __________ __________ _______n__
B/Israel/12/04           __________ __________ __________ __________ _______g__ __________ __________ __________ __________ _______ns_
B/Johannesburg/22/04     __________ __________ __________ __________ _______g__ __________ __________ __________ __________ _______ns_
            Consensus    HRFGETKGNS APLIIREPFI ACGPKECKHF ALTHYAAQPG GYYNGTREDR NKLRHLISVK LGKIPTVENS IFHMAAWSGS ACHDGKEWTY IGVDGPDSNA
```

Figure 12 (continued)

	201									300
B/Harbin/7/94	__________	__________	__________	____d_____	_________k	__________	__________	_e________	__________	__________
B/Yaros1avl/1/02	__________	________k_	__________	____d_____	__________	__________	__________	__________	__________	__________
B/England/23/04	__________	________k_	__________	____d_____	___p______	__________	__________	__________	__________	__________
B/Oslo/71/04	__________	________k_	__________	____d_____	___p______	__________	__________	__________	__________	__________
B/Lyon/GI/72/04	__________	________k_	__________	____d_____	___p______	__________	__________	__________	__________	__________
B/Milan/66/04it	__________	________k_	__________	____d_____	___p______	__________	__________	__________	__________	__________
B/Paris/2113/04	__________	________k_	__________	____d_____	___p______	__________	__________	__________	__________	__________
B/Oslo/586/04nw	__________	________k_	__________	____d_____	___p______	__________	__________	__________	__________	__________
B/Sichuan/379/99	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Shanghai/361/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Jiangsu/10/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Bangkok/460/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Bucharest/795/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Geneva/6159/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Paris/1962/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Hong Kong/64/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Shantou/1212/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Tehran/80/02	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Moscow/3/03	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Madagascar/72134/04	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Geneva/5079/03	__________	__________	__________	__________	_______v__	__________	__________	__________	__________	__________
B/Parma/1/04it	__________	__________	__________	__________	_______v__	__________	__________	__________	__________	__________
B/Israel/12/04	__________	__________	__________	__________	_______v__	__________	__________	__________	__________	__________
B/Johannesburg/22/04	__________	__________	__________	__________	_______v__	__________	__________	__________	__________	__________
Consensus	LLKIKYGEAY	TDTYHSYANN	ILRTQESACN	CIGGNCYLMI	TDGSASGISE	CRFLKIREGR	IIKEIFPTGR	VKHTEECTCG	FASNKTIECA	CRDNSYTAKR

	301									400
B/Harbin/7/94	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Yaros1avl/1/02	__________	__________	________n_	__________	__________	_______e__	__________	__________	__________	_e________
B/England/23/04	__________	__________	________n_	__________	__________	__________	__________	__________	__________	_e________
B/Oslo/71/04	__________	__________	________n_	__________	__________	__________	__________	__________	__________	_e________
B/Lyon/GI/72/04	__________	__________	________n_	__________	__________	__________	__________	__________	__________	_e________
B/Milan/66/04it	__________	__________	________n_	__________	__________	__________	__________	__________	__________	_e________
B/Paris/2113/04	__________	__________	________n_	__________	__________	__________	__________	__________	__________	_e________
B/Oslo/586/04nw	__________	__________	________n_	__________	__________	__________	__________	__________	__________	_e________
B/Sichuan/379/99	__________	__________	__________	__________	__________	__________	__________	__________	__________	__________
B/Shanghai/361/02	__________	__________	__________	__________	_n________	__________	__________	__________	________i_	__________
B/Jiangsu/10/03	__________	__________	__________	__________	_n________	__________	__________	__________	__________	__________
B/Bangkok/460/03	__________	__________	__________	__________	_n________	__________	__________	__________	__________	__________
B/Bucharest/795/03	__________	__________	__________	__________	_n________	__________	__________	__________	__________	__________
B/Geneva/6159/03	__________	__________	__________	__________	_n________	__________	__________	__________	__________	__________
B/Paris/1962/04	__________	__________	__________	__________	_n________	__________	__________	__________	__________	__________
B/Hong Kong/64/04	__________	__________	__________	__________	_n________	__________	__________	__________	________s_	__________
B/Shantou/1212/04	__________	__________	__________	__________	_n________	__________	__________	__________	__________	__________
B/Tehran/80/02	__________	__________	__________	__________	__________	__________	__________	__________	________a_	__________
B/Moscow/3/03	__________	__________	__________	__________	__________	__________	__________	__________	________a_	__________
B/Madagascar/72134/04	__________	__________	__________	__________	__________	__________	__________	__________	________a_	__________
B/Geneva/5079/03	__________	__________	__________	__________	__________	__________	__________	__e_______	________a_	__________
B/Parma/1/04it	__________	__________	__________	__________	__________	__________	__________	__e_______	________a_	__________
B/Israel/12/04	__________	__________	__________	__________	__________	__________	__________	__________	________a_	__________
B/Johannesburg/22/04	__________	__________	__________	__________	__________	__________	__________	__________	________a_	__________
Consensus	PFVKLNVETD	TAEIRLMCTE	TYLDTPRPDD	GSITGPCESN	GDKGSGGIKG	GFVHQRMASK	IGRWYSRTMS	KTKRMGMGLY	VKYDGDPWTD	SDALALSGVM

Figure 12 (continued)

	401						466
B/Harbin/7/94	__________	__________	__________	_____k____	__________	__________	______
B/Yaros1avl/1/02	__________	__________	__________	_____t____	__________	__________	______
B/England/23/04	__________	__________	__________	_____t____	__________	__________	______
B/Oslo/71/04	__________	__________	__________	_____t____	__________	__________	______
B/Lyon/GI/72/04	__________	__________	__________	_____t____	__________	__________	______
B/Milan/66/04it	__________	__________	__________	_____t____	__________	__________	______
B/Paris/2113/04	__________	__________	__________	_____t____	__________	__________	______
B/Oslo/586/04nw	__________	__________	__________	_____t____	__________	__________	______
B/Sichuan/379/99	__________	__________	__________	__________	__________	__________	______
B/Shanghai/361/02	__________	__________	__________	__________	__________	__________	______
B/Jiangsu/10/03	__________	__________	__________	__________	__________	__________	______
B/Bangkok/460/03	__________	__________	__________	__________	__________	__________	______
B/Bucharest/795/03	_p________	__________	__________	__________	__________	__________	______
B/Geneva/6159/03	__________	__________	__________	__________	__________	__________	______
B/Paris/1962/04	__________	__________	__________	__________	__________	__________	______
B/Hong Kong/64/04	__________	__________	__________	__________	__________	__________	______
B/Shantou/1212/04	__i_______	__________	__________	__________	__________	__________	______
B/Tehran/80/02	__________	__________	__________	__________	__________	__________	______
B/Moscow/3/03	__________	__________	__________	__________	__________	__________	______
B/Madagascar/72134/04	__________	__________	__________	__________	__________	__________	______
B/Geneva/5079/03	__________	__________	__________	__________	__________	__________	______
B/Parma/1/04it	_p________	__________	__________	__________	__________	__________	______
B/Israel/12/04	__________	__________	__________	__________	__________	__________	______
B/Johannesburg/22/04	__________	__________	__________	__________	__________	__________	______
Consensus	VSMEEPGWYS	FGFEIKDKKC	DVPCIGIEMV	HDGGKETWHS	AATAIYCLMG	SGQLLWDTVT	GVDMAL